

The Effects of Subsidy Policy on Price and Quality of Child Care

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Abstract

Quality child care can improve low-income children’s outcomes. The largest U.S. child care program gives states \$12.4 billion annually to fund child care subsidy programs targeting low-income families. Given a fixed budget, states trade off serving more families against giving larger subsidies to fewer families. Which choice enrolls more low-income families in quality care is unclear. Using novel administrative data and quasi-experimental variation, I show that higher maximum subsidies increase out-of-pocket payments and increasing the proportion of local families receiving subsidies increases quality of local care. I then estimate a structural model of the child care market with endogenous prices, quality, and number of families subsidized and use it to evaluate Wisconsin’s 2022 reform, which raised the maximum subsidy by 130%. I compare this policy against spending the same budget to serve more families. Price increases explain roughly two-thirds of an 18% decline in formal-care enrollment. Additionally, 16% more eligible children enroll in high-quality formal care, in part because families who would otherwise be subsidized switch to public care. However, 141% more low-income families drop out of formal care than take-up high-quality care.

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1 Introduction

High-quality child care can improve low-income children’s long-run outcomes, including their educational attainment and earnings (Almås, Drange, Meghir and Zachrisson, 2026; Bailey, Sun and Timpe, 2021; Garces, Thomas and Currie, 2002; Gray-Lobe, Pathak and Walters, 2023; Havnes and Mogstad, 2011), as well as their participation in risky or criminal behavior (Anders, Barr and Smith, 2023; Ando, Mori and Yamaguchi, 2025; Carneiro and Ginja, 2014; Garces et al., 2002; Gray-Lobe et al., 2023; Smith, 2015). Access to child care generally has the largest positive effects on children from low-income families because alternative care arrangements are often lower-quality. Indeed, low-quality care can harm children’s cognitive and socio-emotional skills in the short-run (Baker, Gruber and Milligan, 2008; Herbst and Tekin, 2016). Low-quality care may also harm children’s long-run health and criminal behavior (Baker, Gruber and Milligan, 2019).

A common policy to enroll more families in high-quality care is to offer child care subsidies. The largest child care program in the United States, the Child Care and Development Fund (CCDF), gives states \$12.4 billion annually to fund child care subsidy programs (Department of Health and Human Services, 2026). States then must decide how to allocate the funding they have been given. Given limited funding, states must trade off serving more families against giving larger subsidies to fewer families.¹ Quantifying this trade-off is difficult. Both how many families are subsidized and how much each recipient earns affect demand and, in turn, the price and quality of local care may respond in equilibrium. Estimating this trade-off also requires estimating, in equilibrium, how many fewer families are subsidized if the state offers a larger per-family subsidy.

In this paper, I evaluate the enrollment effects of a larger maximum subsidy that Wisconsin implemented in 2022. I compare these enrollment effects to offering a smaller subsidy to more families, while keeping the budget fixed. I determine the extent to which these enrollment

¹Offering subsidies to more families does not require states to change eligibility criteria: states are not obligated to serve all eligible families and often do not have the resources to do so.

changes come predominantly from changes in family choices, prices, or quality.

To motivate a model of the child care market with endogenous prices and quality, I first show that child care providers change their price and quality in response to changes in the proportion of families subsidized locally and the maximum subsidy. Both the proportion of families subsidized and the maximum subsidy are endogenous, so I use two sources of variation to disentangle the effects of each. The first is variation in distance to offices where families can verify their eligibility for subsidies. I show that a greater share of eligible families are subsidized in neighborhoods located closer to these income verification offices. After controlling for the proportion of nearby families eligible for subsidies, the proportion of nearby families using the Supplemental Nutritional Assistance Program, urbanicity, and other controls, I argue that the distance to these offices is plausibly exogenous. Second, I leverage variation in the maximum subsidy over time, by county, and by provider type. While the maximum subsidy is generally higher in counties with more expensive care, I control for these baseline differences and use the fact that the increase in the maximum subsidy was not uniform in timing or magnitude. To evaluate these results, I use novel administrative data from Wisconsin on prices, quality, subsidized enrollment, and other provider characteristics.

Holding the maximum subsidy fixed at the average, I find that subsidizing 1 percentage point (pp) more families increases the likelihood that a given census tract has a high-quality provider by 1.8pp. On the other hand, holding the proportion of families subsidized fixed at the average, I find mixed evidence for the effect of the maximum subsidy on the quality of available care. Under the preferred specification, I find no significant effect of the maximum subsidy on whether a census tract has a high-quality provider.

I also find that out-of-pocket payments increase significantly more when the maximum subsidy increases. Fixing the proportion of families subsidized at the average, a \$10 per week increase in the maximum subsidy increases out-of-pocket payments by 18%, driven in part by a 6.6pp increase in the probability a subsidized family has to pay anything out-of-pocket.

The sticker price for chosen providers increase by 5.1%.² Fixing the maximum subsidy at the average, increasing the proportion of families subsidized by 1pp increases out-of-pocket payments by an insignificant 1.3pp and the sticker price of chosen providers increase by 0.2%.³

While these quasi-experimental results are suggestive, there remain open questions on the effect of a higher maximum subsidy under a fixed budget. First, it is unclear how many fewer families would be subsidized if the state increases the maximum subsidy. A larger maximum subsidy may change not only families' choice of provider, but also the market price and quality of care. Second, while the quasi-experimental results suggest that offering a larger maximum subsidy to fewer families may affect the price and quality of care available, it is unclear how much these changes affect where and whether low-income families choose to enroll.

With these questions in mind, I develop a model of the child care market with endogenous prices, quality, and number of families subsidized.⁴ Given the number of subsidized families and the maximum subsidy offered, the market equilibrium is determined in three stages. First, providers invest in quality based on anticipated demand and local competition. Second, providers observe their quality and the quality of nearby providers and set prices to maximize expected profits. Third, families choose a private or public provider, or to forgo childcare altogether, based on provider characteristics. Under an exogenous budget and maximum subsidy, the state pulls eligible families from a randomized queue, offering subsidies to as many families as it can afford.⁵ Under this model, three channels may affect the quality of care eligible families receive after the maximum subsidy or the number of families subsidized

²The maximum subsidy is, on average, \$199.53 per week in January 2020.

³As a robustness check, I show that similar patterns hold if I use a difference-in-differences approach comparing the effects of different policy changes in Wisconsin and Pennsylvania between high- and low-poverty neighborhoods.

⁴I assume that Head Start and public providers do not change their price and quality in response to these policies. I do not observe the non-profit status of providers. I account for the possible presence of non-profit providers by incorporating unobserved cost shocks into the model. Providers who set lower prices, perhaps because of their non-profit status, are estimated to have lower unobserved costs.

⁵Families are not obligated to use the subsidy if they are offered one.

change.

First, holding price, quality, and other provider characteristics fixed, families may choose different providers (the *sorting* channel). Subsidized families pay lower out-of-pocket payments than unsubsidized families. In addition, subsidized families pay a smaller difference in their out-of-pocket payment across different providers than unsubsidized families. Subsidizing more families then increases demand for desirable, non-price characteristics, like quality. Increasing the maximum subsidy further reduces the difference in out-of-pocket payments across providers, further increasing demand for non-price characteristics. When both the maximum subsidy increases and fewer families are subsidized, it is unclear how family choices, conditional on market offerings, will change.

Second, providers may change their prices (the *endogenous price* channel). Providers cannot price discriminate against subsidy recipients, so they set prices based on demand from both subsidized and unsubsidized families. Subsidized families are price insensitive up to the maximum subsidy. As more families are subsidized, providers with prices below the maximum subsidy have a greater incentive to set their price closer to the maximum subsidy. Some providers may increase their prices more than others. Subsidizing more families or increasing the maximum subsidy increases demand for desirable, non-price characteristics. As high-quality providers obtain a greater market share, they can set higher prices.

Third, providers may change their quality (the *endogenous quality* channel). Providers invest more in quality if offering higher-quality care leads to higher expected profits than offering lower-quality care. Providers earn a greater profit if they can charge a higher mark-up, which is determined by enrollment and how much their market share responds to prices. If demand for high-quality care increases and the cost of providing quality care remain fixed,⁶ providers have greater incentive to invest in quality.

⁶To be sure, it is possible that the cost of offering high-quality care may increase if more providers invest in quality. For example, as demand for child care workers increase, wages may also increase. This is not the focus of this paper, but may reduce the extent to which providers update their quality, and increase the price of care as quality improves. Given the counterfactual results, we may expect that even fewer families enroll in formal or high-quality private care if costs are also endogenous.

Under a fixed budget, the trade-off between offering a higher maximum subsidy versus subsidizing more families depends on how many families lose out on a subsidy offer to pay for the higher maximum-subsidy. To determine how many fewer families are subsidized, I assume the state offers subsidies to as many families as it can afford, treating all eligible families as if they are in a randomized queue. Holding all else fixed, if the state charges a higher maximum subsidy, the state pays more for families who enroll at providers charging prices higher than the prior maximum subsidy. However, providers may choose to raise prices and families may choose more expensive providers, both of which may increase the amount the state has to pay per-family.

I estimate the model using novel data from Wisconsin between 2022 and 2025. During this period, I observe subsidized and total enrollment at almost all licensed providers in Wisconsin.⁷ Because public care is an important alternative source of care, I also rely on geolocated enrollment data for all public 4-K programs, including data by Free and Reduced Price Lunch status.⁸ I also observe state quality ratings at 87% of providers, which are discrete values the state assigns annually based on both structural and process quality.⁹ Because I observe enrollment at unrated providers, I can estimate heterogeneous unobserved demand for these providers. Given the lack of quality rating data on these providers, allowing for this heterogeneity better captures potential substitution patterns. I link price, quality ratings, other licensing characteristics and Head Start status, to addresses. By geolocating providers, I can account for spatial variation in the kinds of families that live near each provider.

I estimate the parameters of demand and the marginal cost function using a two-step

⁷I do not account for the nanny or informal care market. Using 2025 SIPP data, I find that 82% of families with children ages 0-14 paying for non-relative or formal care rely on center, family, or after-school care. In Wisconsin, any provider who supervises at least four children simultaneously outside of the family's own home, excluding schools or public recreational options, must be licensed [DCF \(n.d.\)](#).

⁸Wisconsin school districts use "4-K" to refer to their public preschools available for children under the age of five. In most school districts, preschool is only available to four year olds. For consistency I use the term "4-K" throughout to refer to these kinds of providers.

⁹Structural quality relates to materials and physical space available for children. Process quality captures how teachers interact with children and other aspects of curriculum.

generalized method of moments (GMM) procedure (Berry, Levinsohn and Pakes, 1995; Conlon and Gortmaker, 2020). To address endogeneity of prices and enrollment, I use three sources of variation: differences in nearby provider characteristics, spillovers from a child care worker wage subsidy program that changed over time, and the distance to the nearest income verification office. I also match moments from survey and administrative data to discipline the estimates and identify variation in demand by eligibility status. With demand estimates in hand, I estimate the parameters for the returns to investing in quality by using nested maximum likelihood estimation (MLE), matching the observed quality distribution while ensuring that implied investment is consistent with implied beliefs about other firms' actions.

I find that families who are eligible for subsidies are more sensitive to price and distance from care.¹⁰ Families prefer high-quality care conditional on being quality-rated, but there is significant variation in taste for quality.

I then evaluate a reform Wisconsin implemented in 2022, which increased the maximum subsidy from the 35th percentile of prices to the 80th percentile of prices. I compare the effects of the larger maximum subsidy on enrollment to offering the prior maximum subsidy to more families, under the same budget. I then decompose the effects on enrollment into three channels: changes in sorting (fixing price and quality), endogenous prices, and endogenous quality.

I find that increasing the maximum subsidy reduced enrollment in formal care by 6.3 pp, a decrease of 18%, relative to offering subsidies to more families. 67% of the decline in formal care enrollment is from higher prices. Indeed, prices increase by 9% under the larger maximum subsidy.

Increasing the maximum subsidy also increases quality of care for some children. Because fewer children are subsidized, 13% more children enroll in public 4-K or Head Start. Subsidized children are also 55% more likely to enroll in higher-quality private care. However, because

¹⁰These estimates are preliminary as of September 30, 2026. Please refer to the most up to date version of the paper.

fewer children are subsidized, only 13% more *eligible* children enroll in high-quality private care. Most of the increase in high-quality private care is from increasing the quality of available local care. When weighing the increase in high-quality care against the decrease in formal care though, 141% more children drop out of formal care than switch to high-quality care.

This paper builds on a wider literature evaluating child care subsidy policy on private markets, especially a smaller set using structural methods.¹¹ [Berlinski, Ferreyra, Flabbi and Martin \(2024\)](#) and [Borowsky, Brown, Davis, Gibbs, Herbst, Sojourner, Tekin and Wiswall \(2022\)](#) treat child care as a choice between discrete types of care, including parental care. Parental care then has the cost of preventing mothers from working. Providers are price-takers in both models. [Bodere \(2023\)](#) extends this work by allowing providers to set mark-ups. As in similar work on education markets by [Neilson \(2025\)](#), providers set these mark-ups differently based on which families live nearby. I extend this work by evaluating alternative policies under a program’s budget constraints, when the state may not be able to subsidize all eligible families. The richness of my data allows me to also evaluate the extent to which subsidies may crowd-out public care and to estimate heterogeneous unobserved demand for unrated providers and for public providers.

I also contribute to the literature evaluating interventions to improve quality of child care for low-income children. Many consider interventions to directly improve quality, say by increasing per-child expenditures or teacher training ([Currie and Neidell, 2007](#); [Francis and Barnett, 2019](#); [Joo, Magnuson, Duncan, Schindler, Yoshikawa and Ziol-Guest, 2020](#); [Walters, 2015](#)). I determine how subsidy policy design may crowd-in quality investments from private providers. Other work has shown that children from low-income families are more likely to enroll in low-quality care than children from high-income families, even when universal care is available ([Drange and Telle, 2018](#)) or after receiving more information on the quality of providers ([Herbst, 2018](#)). Subsidized children also tend to enroll in low-quality

¹¹A related paper by [Garcia-Vazquez \(2026\)](#) evaluates the effect of child care regulations on private markets instead of subsidy policies.

care ([Herbst and Tekin, 2016](#)). I demonstrate that availability of quality care is relevant for subsidized family choices.

Finally, I contribute to the broader literature on how subsidy generosity may affect prices, which has been evaluated in other industries such as health ([Alexander and Schnell, 2024](#); [Cabral, Geruso and Mahoney, 2018](#); [Duggan, Starc and Vabson, 2016](#)) and housing ([Collinson and Ganong, 2018](#); [Eriksen and Ross, 2015](#); [Fack, 2006](#)). I demonstrate that a higher maximum subsidy can significantly increase prices. I also show how changing the number of subsidized families may affect the kind of care available on the market.

The rest of the paper is organized as follows. Section 2 reviews the institutional context. Section 3 discusses the data. Section 4 provides some stylized facts to motivate the model, which is introduced in Section 5. Section 6 explains how the model is estimated. Section 7 presents counterfactuals. Section 8 concludes.

2 Institutional Context and Data

2.1 The Child Care Market in Wisconsin

Wisconsin has a diverse child care market, including both private and public options. The majority of enrollees in my sample enroll at private providers.

The majority of private providers are centers. Centers operate from commercial buildings and generally have higher capacities. Private providers that operate from residential facilities (including the provider’s own home) are called family providers. Family providers have a licensed capacity of up to eight children. Because of their low capacity, while family providers represent 46.2% of all licensed providers, only 9% of private enrollees attend a family provider. Family providers are more popular for subsidized families, however, with 17.3% of subsidized children enrolled at family providers.

36.1% of enrollees in my sample attend public care. The largest public option in Wisconsin, public 4-K, is administered by local Wisconsin school districts. The program is free for all

four-year-olds, but is optional. In 2023, 63% of Wisconsin four-year-olds enrolled in 4-K (NIEER, 2023). Similar to other grades, school districts are obligated to provide instruction to all children who live within the school district. 6.6% of public providers are charter schools and 6.7% also offer 3-K (e.g., preschool for three-year-olds). 32.4% have full school-day schedules. Of the remaining schools, 39.0% meet less than the full school week and 57.6% meet less than a full school-day. In addition, Head Start is available to families with incomes under the poverty line. My sample includes Head Start providers who are licensed to offer group care. I exclude, for example, Early Head Start providers which focus on in-home family support services.

2.2 Subsidy Use Across Wisconsin

In Wisconsin, all families earning below 200% of the poverty line are eligible for subsidies.¹² Yet many eligible families fail to receive subsidies. About 17% of families with children in poverty receive child care subsidies. Subsidy use is higher for families with younger children: about 29% of children under age five in poverty are subsidized.¹³ Despite the relatively small size of the program, subsidized children make up a large share of private enrollment. In my sample, 28% of full-time-equivalent enrollment at private providers is from subsidized families. The subsidized share of enrollment is highly heterogeneous over space though. In Figure 1, I map the share of private enrollment in each census tract from subsidized families. In many census tracts, fewer than 5% of all enrollment comes from subsidized children. But in some neighborhoods of urban Milwaukee, over 85% of private enrollment comes from subsidized children.

Comparing Figure 1 with Figure 2, neighborhoods where enrolled children are the most likely to be subsidized are often neighborhoods with more eligible children. Of course, families

¹²Prior to July 2024, families had to earn no more than 185% of the poverty line.

¹³Estimated using ACS 5-year estimated count of families with children under age 18 living in poverty (2024), Wisconsin Department of Children and Families counts of families served under the poverty line (July 2026), and Wisconsin Department of Children and Families counts of children under age five (2nd quarter, 2026). I assume in the estimate for children under age five that families with children under age five participating in the program are similarly likely to be under the poverty line.

eligible for subsidies are more likely to be subsidized, but only a relatively small share of eligible families are subsidized. The high share of private enrollment that are subsidized in these neighborhoods suggests that low-income families are unsubsidized are also less likely to enroll in private care.

Even conditional on the share of families eligible for subsidies, it may be easier to obtain subsidies if you live in certain neighborhoods. Some neighborhoods are located closer to income verification offices, where families can provide the paperwork to demonstrate they meet income requirements. Families must verify their income annually, increasing the costs of not only initially signing up, but also continuing to receive subsidies. Figure 2 shows where each of these offices are located across the state.¹⁴ 56.5% of eligible families live at least 5 miles from the nearest income verification office. In Figure 3, I show, conditional on the proportion of families eligible for subsidies, the relationship between distance to the nearest income verification office and the share of private enrollees who are subsidized. For every mile from the nearest income verification office, the subsidized share of private enrollment falls by 3.5%, controlling for county, the share of families eligible for subsidies, and local urbanicity. Similarly, for every mile from the nearest income verification office, the estimated share of eligible families receiving subsidies falls by 1.7%.

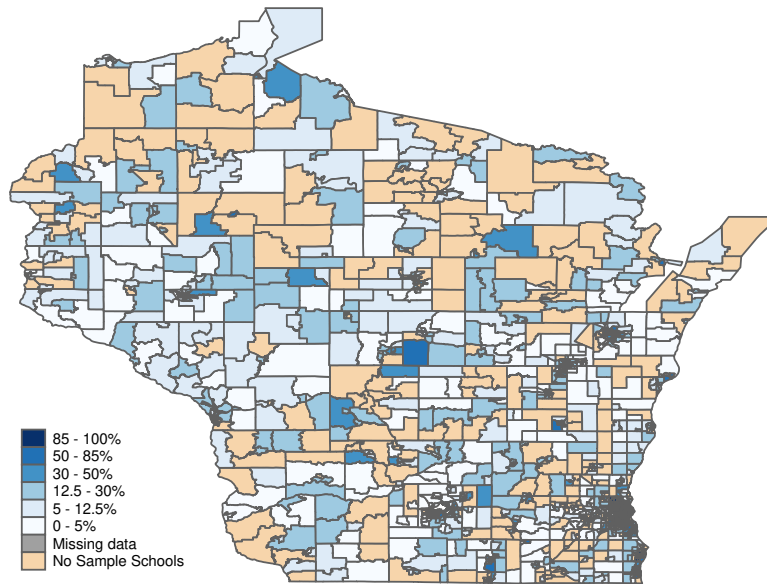
2.3 Wisconsin Subsidy Policy from 2017-2025

Subsidized families can choose from any licensed private provider with a quality rating.¹⁵ Say the family chooses a provider with price p . If the price is below a *maximum subsidy*¹⁶, the

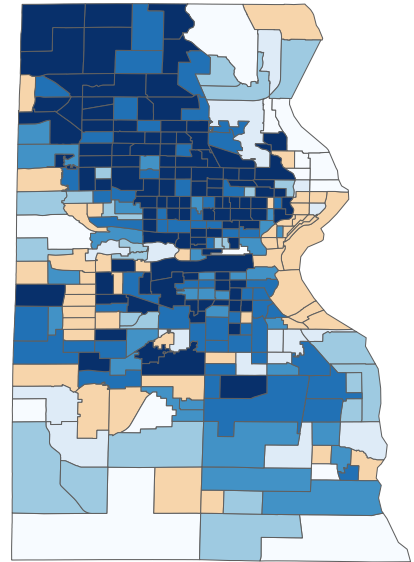
¹⁴Families cannot necessarily go to the office that is truly nearest to them – families must go to offices within their “region.” While these regions are often large enough that the nearest income verification office is within the family’s region, this requirement increases the average miles eligible families travel to their nearest office by 5%. While the state allows families to also apply online, low-income families in Wisconsin have historically been more likely to apply for other programs in-person. Families are also more likely to have a successful application if they apply in person (Voskuil, Friedsam and DeLeire, 2011). Others have leveraged office closings to causally identify the effect of living near income verification offices. These studies have found persistent negative effects on participation in SSDI/DI (Deshpande and Li, 2019) and SNAP (Bitler, Cook, East, Porter and Tiehen, 2025).

¹⁵All providers which are quality rated must agree to accept subsidies.

¹⁶Also referred to as a maximum reimbursement rate.

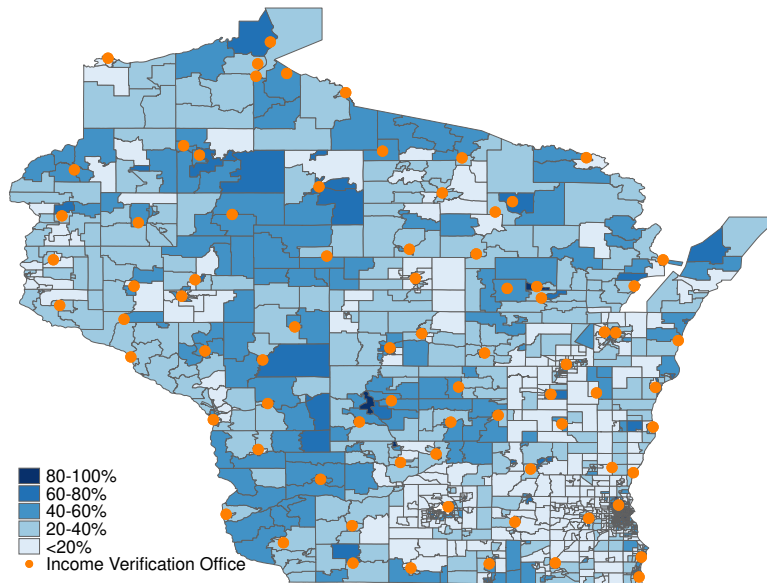


(a) Wisconsin

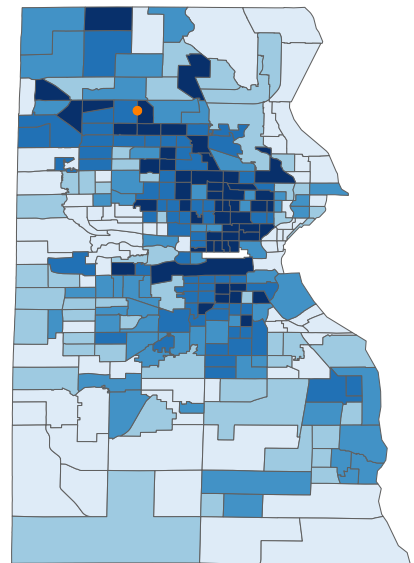


(b) Milwaukee

Figure 1: Proportion of Enrolled Children with Subsidies by Census Tract (2022-2025)



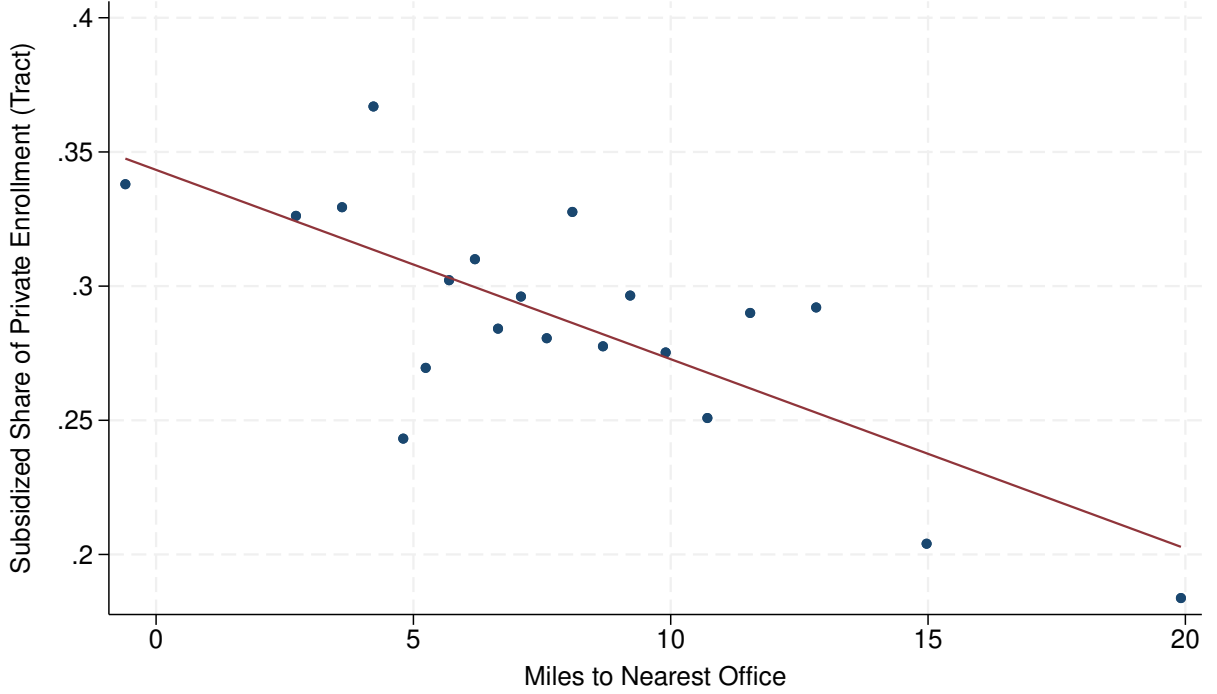
(a) Wisconsin



(b) Milwaukee

Figure 2: Proportion of Families Eligible for Subsidies (2025)

Figure 3: Providers Near Income Verification Offices Have More Subsidized Enrollees



Outcome is the proportion of enrollment in the census tract from subsidized families (2022-2025). Controls for county, urbanicity, and the proportion of families eligible for subsidies.

state covers the cost of care. However, if p is greater than the maximum subsidy $MaxSub$, the family pays the difference. Out-of-pocket payment can then be defined as:

$$OOP(p; MaxSub) = \begin{cases} p - MaxSub & \text{if } p > MaxSub \\ 0 & \text{if } p \leq MaxSub \end{cases} \quad (1)$$

Wisconsin sets the maximum subsidy separately by the age of children served, provider type (e.g., center or family provider), and county. Before 2017, the maximum subsidy had previously been set in 2014 to match local prices. Generally, maximum subsidies are higher for centers serving younger children. Maximum subsidies are higher in counties with higher local prices. In 2017, outside of Tribal Nations,¹⁷ the maximum subsidy was as low as \$99 a week for family providers caring for school-aged children in rural Jackson County and as

¹⁷Due to limited ACS data, I exclude Tribal Nations from the sample.

high as \$250 a week for centers caring for infants in Dane County (i.e., home to Madison).

In March 2018, Congress passed a 38% increase in CCDF funding ([DHS, 2018](#)). Thereafter, Wisconsin increased its maximum subsidy in a staggered fashion from October 2018-October 2019. The roll-out varied by ages served, provider type, and county. Averaging across county, the roll-out is summarized in [Figure 4](#). In October 2018, Wisconsin increased the maximum subsidy for children ages zero to three by 5% or \$175 per week, whichever was higher. Wisconsin then increased the maximum subsidy by an additional 5% in January 2019.¹⁸ Finally, Wisconsin increased its maximum subsidy again in October 2019 to the 35th percentile of prices.¹⁹ If the state determined the 35th percentile of prices was lower than the maximum subsidy, the maximum subsidy remained the same. The percent increase in the maximum subsidy from 2017 to October 2019 ranged from as low as 5% to as high as 77%.²⁰

Despite the increase in federal funding, the number of subsidized families fell by 12% from 2017 to 2020. Eligibility cut-offs did not change over this period. However, the allocation of funds shifted over this period. The average subsidy paid per family increased, in real terms, from \$845 per month in fiscal year 2016 to \$1,217 per month in fiscal year 2019, a 44% increase. Funding for administration of the program, including for determining the eligibility of applicants, fell statewide by 2.1% over the same period ([Gentry, 2017,1](#))

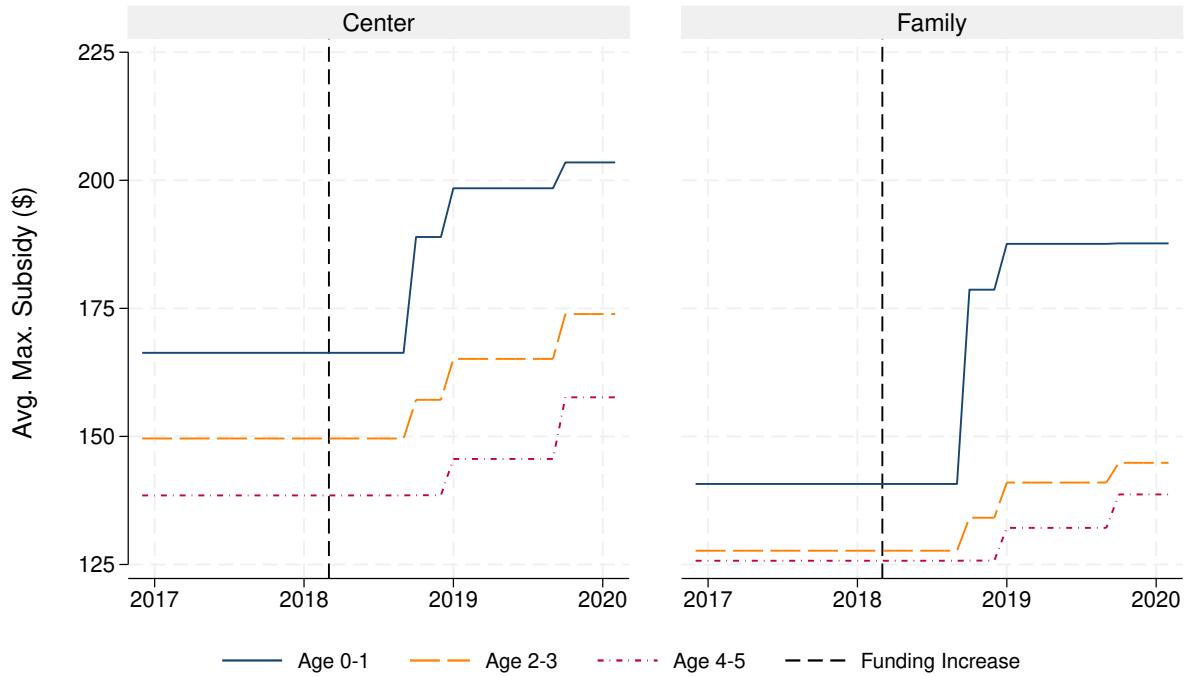
Excluding supplementary and one-time funding increases, Congress increased CCDF funding by 15% between the 2019 and 2021 fiscal years. Under the American Rescue Plan

¹⁸In practice, because of the \$175 a week floor, some providers had very large initial increases in their maximum subsidy, especially family providers. For family providers serving infants in Jackson County for example, the maximum subsidy increased by 49%.

¹⁹Based on prices submitted in a survey of providers. For counties or provider types with few providers, Wisconsin grouped similar counties together, based on the urban proportion of each county.

²⁰Prior to 2022, Wisconsin also set its maximum subsidy differently by quality rating, offering an “add-on” to providers with higher quality ratings, proportional to the maximum subsidy. From October 2018 to January 2019, Wisconsin decreased its add-on for three-star providers by 3.8%, increased its add-on for four-star providers by 0.9% and increased its add-on for five-star providers by 1.6%. While the changes to the add-ons are small relative to the increase in the maximum subsidy, because the add-ons are proportional to the maximum subsidy, the stylized facts may conflate the role of these add-on changes and increases in the maximum subsidy. From 2022-2025, Wisconsin had no such add-on system, so I use the structural model to evaluate the effect of changes in the maximum subsidy on quality absent these add-ons.

Figure 4: Maximum Subsidy Increases from October 2018 to October 2019



Nominal average maximum subsidy weighted by county population of eligible families. Funding increase shows the timing of passage of Congressional funding increase. Centers operate from commercial facilities. Family providers operate from residential facilities. Rates by age of participating child shown.

Act and the Coronavirus Response and Relief Supplemental Appropriations, Congress also allocated an additional \$48.965 billion (83% of allocations for FY2021), which states were obligated to spend by 2024 ([Lynch and Boyle, 2023](#)). With this large funding increase, Wisconsin increased its maximum subsidy in January 2022, to the 80th percentile of prices. Providers in my sample faced an average increase in the maximum subsidy of 130%.²¹ The average subsidy paid per family in fiscal year 2023 was \$1,666 per month ([Gentry, 2025](#)), representing at least half of monthly income for 79% of participating families.

²¹Details can be found at [DCF \(2026\)](#). Average increase is capacity-weighted and based on age-adjusted prices.

2.4 Data

Most of the data are from Wisconsin’s Department of Children and Families (DCF). Additional data on public 4-K providers comes from Wisconsin’s Department of Public Instruction (DPI).

Quasi-Experimental Sample The IV-DID approach I use in the following section requires I observe prices and quality prior to the maximum subsidy increase. The maximum subsidy increase in January 2022 is missing a reasonable pre-period: the COVID-19 pandemic greatly limited child care use from March 2020 to early 2021. For the IV-DID approach, I therefore evaluate the maximum subsidy increase from October 2018 to October 2019 using data from April 2017 until January 2020.²² I limit the sample to licensed, year-round, full-time providers caring for at least some children younger than five. Because the quasi-experimental results focus on quality and price changes from private providers, I narrow the sample to private providers who accept subsidies (87% of all providers).²³

Structural Model Sample To estimate the structural model, I need data on total enrollment. I only observe total enrollment starting in fall 2020. To avoid estimating the model with data from the pandemic, I estimate the structural model using data from January 2022 to January 2025. I expand the sample to include Public 4-K, Head Start, and other public fee-based programs.²⁴ After dropping providers with missing data, the sample represents 84% of reported total enrollment and 88% of subsidized enrollment.²⁵

²²April 2017 is when the state began its roll-out of Electronic Benefit Transfer cards for its child care subsidy program.

²³Providers in Wisconsin sign an agreement with the state before accepting subsidies. All quality-rated providers must sign an agreement with the state to accept subsidies.

²⁴Public programs are less likely to be available year-round or for the full-day. I keep public providers operating in January and offering care to children younger than five.

²⁵Some providers choose not to report any total enrollment. 13.6% of seats at otherwise in-sample licensed providers are unreported. Subsidized enrollment, however, is observed at every provider. Excluding public providers or Head Starts, sample providers represent 82% of reported total enrollment and 82% of subsidized enrollment. See ?? and Table 10 for a detailed comparison between sample providers and all licensed providers, as well as sample statistics.

Public 4-K Data From DPI, I observe total enrollment by grade level each September,²⁶ including by eligibility for free or reduced-price lunch (e.g., families earn no more than 185% of the poverty line). DPI also collect school addresses and schedules.

Licensing Data As part of licensing, DCF tracks provider addresses, hours and months the provider is licensed to operate,²⁷ ages served, license start and end dates, the type of facility the provider operates from, and licensed capacity. Licensed capacity only reflects building size, rather than minimum student-teacher ratios, and should be treated as an upper-bound for true capacity.

Price and Subsidized Enrollment DCF collects prices and subsidized enrollment in the process of administering subsidies. Providers must submit prices so the state can allocate the correct amount each month. Monthly subsidized enrollment is tracked in the same process.²⁸ Providers can update prices at any time, and prices may vary by the child’s age.²⁹ For the structural model, only 2% of estimated enrollment is dropped from the sample because the provider never submits prices.³⁰

Quality The state’s quality rating system (YoungStar) gives licensed providers a discrete rating from one to five. The quality rating is updated annually. The vast majority of providers (87%) agree to be rated. Providers that apply but fail to meet licensing requirements receive one star, though this is rare.³¹ Any provider that meets licensing standards and is willing to accept subsidies earns two stars. For 2022-2025, 41% of licensed providers have

²⁶I assume January enrollment is the same as September enrollment.

²⁷Licensed hours are not necessarily the same as true operating hours, but rather when the provider can legally operate. I only use licensed hours to classify providers as full- or part-time.

²⁸Wisconsin data comes from records obtained in the process of administering subsidies using using Electronic Benefit Transfer cards, which Wisconsin began using in 2017 (DECE, 2018).

²⁹I do not observe subsidized enrollment by age, so for the stylized facts, I use a simple average over the expected out-of-pocket payments for ages zero to four, and weight the results by total subsidized enrollment. For the structural model, I instead use a fixed-effects model to estimate each provider’s underlying price, differencing out age effects.

³⁰For the structural model, I impute prices in earlier years based on later observations.

³¹In 2022, for example, only four providers in the sample received one star.

two stars. To earn three stars, providers must meet standards for process quality (e.g., how to interact with children), structural quality (e.g., materials and space), and staff education and training. Earning four or five stars requires meeting additional standards for process and structural quality, as well as staff education and training. Providers must also earn an average score of at least four or five, respectively, on the Environment Rating Scale (i.e., ITERS, ECERS or FCCERS), a long-standing standardized assessment ([Blau and Currie, 2006](#)). 28% of Wisconsin providers are at least four stars. We may worry that providers who are high-quality may not be quality-rated because of fixed costs of applying. Nevertheless, I find that unrated providers have higher average student-teacher ratios. Additionally, a validation study conducted in Wisconsin find that lower-rated providers score lower, on average, on the Environment Rating Scale. Additionally, differences in quality between two to three stars is larger than the difference in quality from three to four stars ([Magnuson and Lin, 2015](#)).

Total Enrollment Data Wisconsin offered grants (Child Care Counts) to providers from 2020-2025 based on their total enrollment and employment. Participation was high: only 13.6% of seats at otherwise in-sample licensed providers are unreported. Any licensed provider in good standing could apply.

Additional Survey Data Neighborhood characteristics come from the American Community Survey. For the quasi-experimental results, I use tract-level data. For the structural model, I use block-level data. When necessary, I impute block-level variation from tract-level data. I also match moments on distance traveled and age of enrollment by income based on data from the National Survey of Early Care and Education (2024).

3 Quasi-Experimental Evidence

The model discussed in [section 4](#) assumes providers respond to changes in the number of families subsidized and the changes in the maximum subsidy. In this section, I evaluate whether this assumption is reasonable using the roll-out of the maximum subsidy increase in 2018-2019. I leverage two key sources of variation. The first is variation in distance to income verification offices. The second is variation in the maximum subsidy over time, by county, and by provider type. I build towards a 2SLS-DiD approach in four steps. First, I evaluate the effect of subsidizing more local families in isolation by instrumenting for the proportion of families subsidized with the distance to the income verification office. Second, I evaluate the effect of the maximum subsidy in isolation using a difference-in-differences approach, taking the average maximum subsidy in 2019 to 2020³² as exogenous *conditional* on baseline differences between different counties and provider types. Third, I account for potential correlation between the maximum subsidy and how many families are subsidized, as well as possible interaction effects, using a triple-differences approach. Here, I extend the difference-in-differences analysis from the previous step evaluating the maximum subsidy alone. I also take the *change* in the proportion of families subsidized as exogenous conditional on baseline differences. To make this assumption, I leverage the fact that neighborhoods closer to income verification offices were less exposed to statewide declines in subsidy coverage. A problem with the triple differences approach is that changes in the proportion of families subsidized may also reflect mean reversion. Providers may be less responsive to these temporary fluctuations in the proportion of families subsidized. I therefore apply the 2SLS-DID approach, which instruments for the proportion of local families receiving subsidies with distance to income verification offices. I further control for county, provider type, and year fixed-effects, as well as other local and provider controls, to identify the effects of the proportion of families subsidized and the maximum subsidy on

³²For tract-level quality results, I use the average maximum subsidy faced by all providers. For provider-level out-of-pocket payment results, I use the baseline maximum subsidy faced by that provider.

quality and price of care.

I focus on two outcomes. The first is whether care of a certain quality rating is available in a given census tract. I consider the effects on having care of at least three, four, and five stars.³³ The second is the out-of-pocket payment that subsidized families pay.³⁴

Throughout, I control for local demographics, including measures of the income distribution, marital status for mothers of young children, birth rates by marital and educational status, Supplementary Nutritional Assistance Program (SNAP) receipt, the number of young children who live nearby, and urbanicity. For the out-of-pocket payment results, I further control for the provider’s quality rating and whether it operates from a residential facility. Because I do not observe age, I use age-adjusted out-of-pocket payments and sticker prices.³⁵

3.1 Quasi-Experimental Methods

Both the proportion of families subsidized and the maximum subsidy are endogenous. Neighborhoods with more subsidized families may have higher unobserved demand for formal care, for example. The maximum subsidy is set higher in counties or for provider types with higher prices. In this section, I discuss each approach I use to address this endogeneity.

First, I isolate the effect of subsidizing more families on out-of-pocket payments and quality using an IV approach. Let Y_{it} be the relevant outcome observed in year t for either census tract i or provider i . To address potential endogeneity in the proportion of families subsidized, I instrument with distance to the income verification office. We may worry that the state may establish offices in neighborhoods where families have greater interest in child

³³Five stars is the highest possible rating. Validation studies have suggested that three star providers are more different from two star providers than four star providers. Four and five star providers must receive a certain average rating on the Environmental Rating Scale and three star providers must implement certain curricula and meet staffing standards.

³⁴I observe the out-of-pocket payment and subsidized enrollment at the provider-level. In practice, I use the out-of-pocket payment as the outcome variable for a given provider and weight by subsidized enrollment. I weight all tract-level results by the number of children under age five living in the census tract.

³⁵I observe the price that the provider offers for each age group, as well as the maximum subsidy by age group. To age-adjust the prices, I use a fixed-effects model assuming that the difference in out-of-pocket payments by age are the same across providers. Prices and out-of-pocket payments are centered at the price and out-of-pocket payment for three-year-olds.

care subsidies. Certainly, offices are more likely to be in poorer neighborhoods or urban areas. However, these offices are often established in locations the city already owns to run other, larger, programs. I argue that, conditional on enrollment in programs like SNAP, distance to these offices can be treated as exogenous. As discussed earlier, distance to these offices has a strong effect on whether families enroll in child care subsidies. To summarize, the first and second stages can be described as

$$PrSub_{it} = \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 X_{it} + \zeta_c + \zeta_t + \nu_{it} \quad (1)$$

$$Y_{it} = \beta_0 + \beta_1 \widehat{PrSub}_{it} + \beta_2 X'_{it} + \eta_c + \eta_t + \varepsilon_{it} \quad (2),$$

where $\zeta_c, \zeta_t, \eta_c, \eta_t$ are county and time fixed-effects, X_{it} are a set of demographic and provider controls, $DistOffice_i$ is the distance to the nearest income verification office, and $PrSub_{it}$ is the proportion of families subsidized in the census tract (or the provider's census tract).³⁶

Second, I identify the effect of the maximum subsidy based on its differential roll-out by county and provider type over time in a difference-in-differences design. I measure treatment as the average maximum subsidy faced by the provider (or the average maximum subsidy faced by all providers within the tract³⁷) across 2019 to 2020 ($MaxSub_{i,19-20}$). This measure summarizes both the differential timing of the maximum subsidy increase and variation in the magnitude of the maximum subsidy. I condition on baseline differences between providers or tracts with higher maximum subsidies in later years, as well as county fixed-effects η_c :

$$Y_{it} = \beta_0 + \beta_1 MaxSub_{i,19-20} + \beta_2 Post_t + \beta_3 (MaxSub_{i,19-20} \times Post_t) + \beta_4 X_{it} + \eta_c + \varepsilon_{it}. \quad (2)$$

Third, I account for correlation between the maximum subsidy and the proportion of families subsidized, including possible interaction effects. I take advantage of the fact

³⁶I do not observe the proportion of families subsidized in the census tract directly. Rather, I use the proportion of subsidies used at providers in the given tract relative to the number of young children who live in the census tract. In later model estimates, I demonstrate that families have a strong taste for providers located near where they live.

³⁷If there are no providers in the tract, I use the county average.

that caseloads fell statewide from 2017 to 2020, but did not fall evenly across the state. Let $\Delta \ln PrSub_i$ denote the change in the log proportion of families subsidized in tract i ³⁸ between 2017–2018 and 2019–2020. I treat $\Delta \ln PrSub_i$ as exogenous conditional on baseline differences and county fixed effects. I also once again treat $MaxSub_{i,1920}$ as exogenous conditional on baseline differences and county fixed effects. The triple-differences approach can be summarized as follows:

$$\begin{aligned}
Y_{it} = & \beta_0 + \beta_1 \Delta \ln PrSub_i + \beta_2 MaxSub_{i,19-20} + \beta_3 Post_t \\
& + \beta_4 (\Delta \ln PrSub_i \times MaxSub_{i,1920}) + \beta_5 (\Delta \ln PrSub_i \times Post_t) + \beta_6 (MaxSub_{i,1920} \times Post_t) \\
& + \beta_7 (\Delta \ln PrSub_i \times MaxSub_{i,1920} \times Post_t) + \beta_8 X_{it} + \eta_c + \varepsilon_{it},
\end{aligned} \tag{3}$$

where the relevant outcomes of interest are β_5 , β_6 , and β_7 .

Fourth, I address the possibility that even controlling for baseline differences, the change in the proportion of families subsidized may be endogenous. For example, we may worry that tracts with greater changes in the proportion of families subsidized are more likely to reflect temporary fluctuations in the proportion of families subsidized. I once again instrument for the proportion of families receiving subsidies with distance to the nearest income verification office. I also instrument for the interaction between the maximum subsidy and the proportion of families subsidized with the interaction between the maximum subsidy and distance to the nearest income verification office. I again assume that the maximum subsidy is exogenous conditional on county, year, and provider type fixed-effects. In summary:

$$Y_{it} = \beta_0 + \beta_1 \widehat{PrSub}_{it} + \beta_2 (\widehat{PrSub}_{it} \times \widehat{MaxSub}_{it}) + \beta_3 MaxSub_{it} + \beta_4 X'_{it} + \eta_c + \eta_t + \varepsilon_{it}, \tag{4}$$

³⁸Or the tract that provider i is located in, in the case of out-of-pocket payment outcomes.

where the first-stage equations are

$$\begin{aligned}
PrSub_{it} &= \alpha_0 + \alpha_1 DistOffice_i + \alpha_2 (DistOffice_i \times MaxSub_{it}) + \alpha_3 MaxSub_{it} + \alpha_4 X_{it} \\
&\quad + \zeta_c + \zeta_t + \nu_{1,it} \\
PrSub_{it} \times MaxSub_{it} &= \phi_0 + \phi_1 DistOffice_i + \phi_2 (DistOffice_i \times MaxSub_{it}) + \phi_3 MaxSub_{it} \\
&\quad + \phi_4 X_{it} + \eta_c + \eta_t + \nu_{2,it}.
\end{aligned}$$

3.2 Quality

[Table 1](#) reports the results from the IV approach evaluating only the effect of subsidizing more local families. The instrument is relevant, with a first-stage with an $F = 37.3$. I find that a 1pp increase in the proportion of families subsidized increases the likelihood a given tract has a provider with at least three stars by 2.3pp and increases the likelihood a given tract has a provider with at least four stars by 1.1pp. Both of these estimates are significant at $p < 0.01$. The effect on the likelihood a census tract has at least one five star provider is 0.8pp, an insignificant effect. These results suggest that increasing the number of subsidized families has the greatest effect on increasing access to middling-quality providers.

[Table 2](#) then reports the effect of the maximum subsidy using difference-in-differences. A \$10 higher maximum subsidy increases the likelihood a census tract has at least a three-star provider by 0.5pp ($p < 0.05$), the likelihood of at least a four-star provider by 0.6pp ($p < 0.01$), and the likelihood of at least one five-star provider by an insignificant 0.2pp. These results suggest that increasing the maximum subsidy has the greatest effect on increasing access to middling-quality providers.

[Table 3](#) reports the results from the triple-differences approach. A 10% increase in the proportion of families subsidized leads to a 1pp increase in the probability that a census tract has a provider with at least three stars. The proportion of families subsidized has no significant effect on whether there are providers with higher quality-ratings. Offering a \$10 higher maximum subsidy increases the probability that a census tract has a provider with

Table 1: Higher Local Subsidy Use Increases Access to Middling-Quality Providers

	(1) OLS Any 3+	(2) IV Any 3+	(3) OLS Any 4+	(4) IV Any 4+	(5) OLS Any 5	(6) IV Any 5
Pr. Fam. Sub.	0.012*** (0.001)	0.023*** (0.007)	0.008*** (0.001)	0.011* (0.007)	0.006*** (0.001)	0.008 (0.006)
N	5326	5326	5326	5326	5326	5326
F		37.318		37.318		37.318

Outcome is an indicator for whether the census tract has at least one provider with a YoungStar rating of 3+ (columns 1–2), 4+ (columns 3–4), or 5 (columns 5–6) stars. The sample is Wisconsin census tracts, 2017–2020. Result measured in January of each year. Pr. Fam. Sub. is the proportion of families subsidized ($PrSub_{it}$, 1pp units, mean-centered). IV columns instrument with the minimum distance from the tract to the nearest income verification office. Controls include the number of young children, the proportion of young children’s parents who are married, birth rate to married and college-educated women, coarse income distribution for families with young children (100%, 200%, 300%, 400%, 500+% of the federal poverty line), proportion of families with young children receiving SNAP, tract urbanicity, county and year fixed effects. Observations are weighted by the number of young children per tract. Standard errors are clustered at the tract level and shown in parentheses. F is the Angrist-Pischke conditional first-stage F -statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Higher Maximum Subsidy Increases Access to Middling-Quality Providers

	(1) Any 3+ Star	(2) Any 4+ Star	(3) Any 5 Star
Max. Sub. (\$10/week) $\times$ Post	0.005** (0.002)	0.006*** (0.002)	0.002 (0.003)
N	5326	5326	5326

Outcome is an indicator for whether the census tract has at least one provider with a YoungStar rating of 3+ (column 1), 4+ (column 2), or 5 (column 3) stars. The sample is Wisconsin census tracts, 2017–2020. Result measured in January of each year. Max. Sub. is the 2019–2020 average maximum subsidy ($MaxSub_{i,19-20}$, \$10/week units, mean-centered). Post is an indicator for years ≥ 2019 . Controls include the number of young children, the proportion of young children’s parents who are married, birth rate to married and college-educated women, coarse income distribution for families with young children (100%, 200%, 300%, 400%, 500+% of the federal poverty line), proportion of families with young children receiving SNAP, tract urbanicity, and county fixed effects. Observations are weighted by the number of young children per tract. Standard errors are clustered at the county-year level and shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

at least three stars by 0.4pp ($p < 0.05$) and at least four stars by 0.5pp ($p < 0.01$). There is no significant interaction effect between the two policies.

Table 3: Increasing the Maximum Subsidy or Proportion of Families Subsidized Increases Access to Middling-Quality Providers

	(1) Any 3+ Star	(2) Any 4+ Star	(3) Any 5 Star
Change in Log Prop. Fam. Sub. $\times$ Post	0.099*** (0.030)	0.025 (0.015)	0.014 (0.013)
Max. Sub. (\$10/week) $\times$ Post	0.004** (0.002)	0.005*** (0.002)	0.002 (0.003)
Change in Prop. Fam. Sub. $\times$ Max. Sub. $\times$ Post	0.007 (0.008)	-0.005 (0.004)	0.003 (0.005)
N	5324	5324	5324

Outcome is an indicator for whether the census tract has at least one provider with a YoungStar rating of 3+ (column 1), 4+ (column 2), or 5 (column 3) stars. The sample is Wisconsin census tracts, 2017–2020. Result measured in January of each year. Change in Log Prop. Fam. Sub. is the log change in the proportion of families subsidized between 2017–2018 and 2019–2020 ($\Delta \ln PrSub_i$). Max. Sub. is the 2019–2020 average maximum subsidy ($MaxSub_{i,19-20}$, \$10/week units, mean-centered). Post is an indicator for years ≥ 2019 ($Post_t$). The regression includes all main effects and two- and three-way interactions of Log Change, Max. Sub., and Post; only the interactions with Post are shown. Controls include the number of young children, the proportion of young children’s parents who are married, birth rate to married and college-educated women, coarse income distribution for families with young children (100%, 200%, 300%, 400%, 500+% of the federal poverty line), proportion of families with young children receiving SNAP, tract urbanicity, and county fixed effects. Observations are weighted by the number of young children per tract. Standard errors are clustered at the county-year level and shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 reports the 2SLS-DiD estimates. A 1pp increase in the proportion of families subsidized increases the likelihood a tract has a provider of at least three stars by 4.4pp, a provider of at least four stars by 1.8pp, and a provider of at least five stars by 1.3pp. On the other hand, a \$10 increase in the maximum subsidy has no significant positive effect on the quality of available providers. If anything, the effect of a higher maximum subsidy on having a provider of at least three stars is marginally negative.

The evidence suggests that increasing access to subsidies has a significant effect on whether providers are of at least middling-quality. In the preferred specification, the proportion of families receiving subsidies has a significant effect on whether a census tract has a provider of the highest quality rating, although the effect is smaller than the effect on middling-quality

Table 4: Increasing Access to Subsidies Increases Access to High-Quality Providers

	(1) Any 3+ Star	(2) Any 4+ Star	(3) Any 5 Star
Proportion Families Subsidized (1pp)	0.044*** (0.009)	0.018*** (0.007)	0.013** (0.005)
Max. Sub. (\$10/week)	-0.050* (0.027)	-0.001 (0.018)	0.011 (0.014)
Prop. Fam. Sub. $\times$ Max. Sub.	-0.007*** (0.002)	-0.002 (0.001)	-0.001 (0.001)
N	5319	5319	5319
F (Prop. Fam. Sub.)	35.995	35.995	35.995
F (Prop. Fam. Sub. $\times$ Max. Sub.)	34.721	34.721	34.721

Outcome is an indicator for whether the census tract has at least one provider with a YoungStar rating of 3+ (column 1), 4+ (column 2), or 5 (column 3) stars. The sample is Wisconsin census tracts, 2017–2020. Result measured in January of each year. Proportion Families Subsidized is the proportion of families subsidized ($PrSub_{it}$, 1pp units, mean-centered). Max. Sub. is the year-varying maximum subsidy ($MaxSub_{it}$, \$10/week units, mean-centered). The proportion of families subsidized and its interaction with the maximum subsidy instrumented with the minimum distance from the tract to the nearest income verification office and its interaction with the maximum subsidy. The maximum subsidy main effect is assumed to be exogenous conditional on county, year fixed effects. Controls include the number of young children, the proportion of young children’s parents who are married, birth rate to married and college-educated women, coarse income distribution for families with young children (100%, 200%, 300%, 400%, 500+% of the federal poverty line), proportion of families with young children receiving SNAP, tract urbanicity, county and year fixed effects. Observations are weighted by the number of young children per tract. Standard errors are clustered at the county-year level and shown in parentheses. The F rows report the Angrist-Pischke conditional first-stage F -statistic for each endogenous regressor. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

care. The evidence on the effect of a higher maximum subsidy is more mixed. In the preferred specification, the maximum subsidy has a marginally significant negative effect on whether a census tract has a middling-quality provider and a small, but imprecise positive effect on whether a census tract has a top-quality provider.

3.3 Out-of-Pocket Payments

Table 5 applies the IV approach to provider-level payment outcomes. The first-stage has an $F = 29.295$, suggesting the instrument is relevant. The proportion of families subsidized has no significant effect on out-of-pocket payments, conditional on the provider’s quality.

Table 5: Proportion of Families Subsidized Has Insignificant Effect on Out-of-Pocket Payments

	(1) OLS ln(OOP)	(2) IV ln(OOP)	(3) OLS OOP>0	(4) IV OOP>0	(5) OLS ln(Price)	(6) IV ln(Price)
Pr. Fam. Sub.	0.002 (0.002)	0.002 (0.008)	0.000 (0.001)	0.002 (0.002)	0.000* (0.000)	0.001 (0.001)
N	7560	7560	7560	7560	7560	7560
F		29.295		29.295		29.295

Outcomes are the log age-adjusted out-of-pocket payment for a subsidized family at the provider (ln(OOP), columns 1–2), an indicator for a positive out-of-pocket payment (OOP>0, columns 3–4), and the log sticker price (ln(Price), columns 5–6). Outcomes are at the provider-year level, weighted by subsidized enrollment. The sample is Wisconsin licensed private providers with at least one subsidized enrollee. Result measured in January of each year. Pr. Fam. Sub. is the proportion of families subsidized in the provider’s tract ($PrSub_{it}$, 1pp units, mean-centered). IV columns instrument with the minimum distance from the provider to the nearest income verification office. Controls include the number of young children in the provider’s tract, the proportion of young children’s parents who are married, birth rate to married and college-educated women, coarse income distribution for families with young children (100%, 200%, 300%, 400%, 500+% of the federal poverty line), proportion of families with young children receiving SNAP, tract urbanicity, and star, county, provider-type, and year fixed effects. Standard errors are clustered at the tract-level and shown in parentheses. F is the Angrist-Pischke conditional first-stage F -statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 shows the difference-in-differences estimates of the effect of the maximum subsidy. A \$10 higher weekly maximum subsidy raises out-of-pocket payments by 3.3% ($p < 0.01$) and the likelihood a subsidized family pays anything out-of-pocket by 1pp ($p < 0.05$). However, the effect on the sticker price families pay is much smaller. One possibility is that price

increases are relatively small and concentrated at providers setting their prices near the maximum subsidy.

Table 6: Out-of-Pocket Payments Increase with the Maximum Subsidy

	(1) ln(OOP)	(2) OOP>0	(3) ln(Price)
Max. Sub. (\$10/week) $\times$ Post	0.033*** (0.010)	0.010** (0.004)	0.001 (0.001)
N	7557	7557	7557

Outcomes are the log age-adjusted out-of-pocket payment for a subsidized family at the provider (ln(OOP), columns 1–2), an indicator for a positive out-of-pocket payment (OOP>0, columns 3–4), and the log sticker price (ln(Price), columns 5–6). Outcomes are at the provider-year level, weighted by subsidized enrollment. The sample is Wisconsin licensed private providers with at least one subsidized enrollee. Result measured in January of each year. Max. Sub. is the 2019–2020 average maximum subsidy ($MaxSub_{i,19-20}$, \$10/week units, mean-centered). Post is an indicator for years ≥ 2019 . Controls include the number of young children in the provider’s tract, the proportion of young children’s parents who are married, birth rate to married and college-educated women, coarse income distribution for families with young children (100%, 200%, 300%, 400%, 500+% of the federal poverty line), proportion of families with young children receiving SNAP, tract urbanicity, and star, county, and provider-type fixed effects. Standard errors are clustered at the county-year level and shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7 adds the effect of changes in the local proportion of families subsidized to the difference-in-differences from Table 6. The maximum-subsidy pass-through survives essentially unchanged in magnitude, though the effect on prices (0.1% increase for every \$10 increase in the maximum subsidy) is measured more precisely. The change in the local proportion of families subsidized has no significant effect on out-of-pocket payments.

Table 8 reports the results from the 2SLS-DiD approach, the preferred specification. Both instruments are relevant, with first-stage F -statistics of 34.4 and 42.0. I find that a \$10/week higher maximum subsidy raises out-of-pocket payments by 18.1%, increases the likelihood a subsidized family pays anything out of pocket by 6.6pp, and raises sticker prices by 5.1%. A 1pp higher proportion of families subsidized increases sticker prices by a marginally significant 0.2%. These results suggest, conditional on provider quality, a higher maximum subsidy increases prices and out-of-pocket payments, while subsidizing more families has a relatively small effect.

Table 7: Out-of-Pocket Payments Increase with the Maximum Subsidy

	(1) ln(OOP)	(2) OOP>0	(3) ln(Price)
Log Change in Prop. Fam. Sub. $\times$ Post	-0.107 (0.130)	-0.013 (0.045)	-0.001 (0.008)
Max. Sub. (\$10/week) $\times$ Post	0.038*** (0.012)	0.011*** (0.004)	0.001** (0.001)
Log Change $\times$ Max. Sub. $\times$ Post	-0.029 (0.020)	-0.015** (0.007)	-0.002 (0.002)
N	7556	7556	7556

Outcomes are the log expected out-of-pocket payment for a subsidized family at the provider (ln(OOP), column 1), an indicator for a positive out-of-pocket payment (OOP>0, column 2), and the log sticker price (ln(Price), column 3), all measured at the provider-year level. The sample is Wisconsin licensed private providers with at least two stars, 2017–2020. Result measured in January of each year. Log Change in Prop. Fam. Sub. is the log change in the proportion of families subsidized in the provider’s tract between 2017–2018 and 2019–2020 ($\Delta \ln PrSub_i$). Max. Sub. is the 2019–2020 average maximum subsidy ($MaxSub_{i,19-20}$, \$10/week units, mean-centered). Post is an indicator for years ≥ 2019 ($Post_t$). The regression includes all main effects and two- and three-way interactions of Log Change, Max. Sub., and Post; only the interactions with Post are shown. Controls include the number of young children in the provider’s tract, the proportion of young children’s parents who are married, birth rate to married and college-educated women, coarse income distribution for families with young children (100%, 200%, 300%, 400%, 500+% of the federal poverty line), proportion of families with young children receiving SNAP, tract urbanicity, and star, county, and provider-type fixed effects. Observations are weighted by provider capacity. Standard errors are clustered at the county-year level and shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Out-of-Pocket Payments Increase with the Maximum Subsidy

	(1) ln(OOP)	(2) OOP>0	(3) ln(Price)
Proportion Families Subsidized (1pp)	0.013 (0.012)	0.006 (0.004)	0.002* (0.001)
Max. Sub. (\$10/week)	0.181** (0.085)	0.066** (0.026)	0.051*** (0.009)
Prop. Fam. Sub. $\times$ Max. Sub.	-0.003 (0.002)	-0.001* (0.001)	-0.000 (0.000)
N	7557	7557	7557
F (Prop. Fam. Sub.)	34.363	34.363	34.363
F (Prop. Fam. Sub. $\times$ Max. Sub.)	41.956	41.956	41.956

Outcomes are the log expected out-of-pocket payment for a subsidized family at the provider (ln(OOP), column 1), an indicator for a positive out-of-pocket payment (OOP>0, column 2), and the log sticker price (ln(Price), column 3), all measured at the provider-year level. The sample is Wisconsin licensed private providers with at least two stars, 2017–2020. Result measured in January of each year. Proportion Families Subsidized is the proportion of families subsidized ($PrSub_{it}$, 1pp units, mean-centered). Max. Sub. is the year-varying maximum subsidy ($MaxSub_{it}$, \$10/week units, mean-centered). The proportion of families subsidized and its interaction with the maximum subsidy instrumented with the minimum distance from the tract to the nearest income verification office and its interaction with the maximum subsidy. The maximum subsidy main effect is assumed to be exogenous conditional on county, year fixed effects. Controls include the number of young children in the provider's tract, the proportion of young children's parents who are married, birth rate to married and college-educated women, coarse income distribution for families with young children (100%, 200%, 300%, 400%, 500+% of the federal poverty line), proportion of families with young children receiving SNAP, tract urbanicity, and star, county, provider-type, and year fixed effects. Observations are weighted by provider capacity. Standard errors are clustered at the county-year and shown in parentheses. The F rows report the Angrist-Pischke conditional first-stage F -statistic for each endogenous regressor. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Taken together, the evidence suggests that the proportion of families subsidized has a small effect on what subsidized families pay, conditional on the quality of care they receive. On the other hand, higher maximum subsidies raise both sticker prices and out-of-pocket payments.

4 Model

The stylized facts establish that increasing the number of children subsidized can improve local quality and that increasing the maximum subsidy can increase out-of-pocket payments. But what drives these changes in quality and prices? How much do changes in price and quality affect where and whether families enroll in child care? How many fewer families can the state subsidize if the maximum subsidy increases? To answer these questions, in this section I describe a model of the child care market. Based on the maximum subsidy and the proportion of families subsidized, the market equilibrium is determined in three stages. First, providers invest in quality, anticipating competition and demand. Second, providers observe their own quality and the quality of other providers in the market and set prices to maximize expected profit. Third, families choose a private or public provider, or no provider at all. Providers solve this game using backwards induction. Accordingly, in this section I first discuss demand, then price-setting behavior, and then quality-setting behavior. After establishing the demand and supply model, I discuss the state's problem. Based on the anticipated market equilibrium, the state offers subsidies to as many eligible families as it can afford, given an exogenously set budget and maximum subsidy. I conclude the section by discussing the implications of the model.

4.1 Demand

Family i consumes utility U_{ijt} if they enroll at provider j in market-year t . Utility depends on the characteristics of provider j (in market-year t) and family i 's demographic-specific

preferences. Provider j is associated with a quality rating $Qual_{jt}$, observable characteristics X_{jt} , and unobservable (to the econometrician) average demand shocks ξ_{jt} . Family i must travel a distance d_{ij} to attend provider j and pays an out-of-pocket payment ($OOP(sub_i, p_{jt}; MaxSub_t)$) that depends on family i 's subsidy status, the price provider j charges and the maximum subsidy in market-year t . Family i may also have an unobserved, idiosyncratic shock to demand for provider j (ϵ_{ijt}). Utility can be summarized as:

$$U_{ijt}(sub_i) = \beta_i^X X_{jt} + \beta_i^Q Qual_{jt} + \beta_i^{oop} OOP(sub_i, p_{jt}; MaxSub_t) + \beta_i^d d_{ijt} + \xi_{jt} + \epsilon_{ijt}.$$

As reflected in the i sub-scripts on the β terms, preferences may vary by i , either based on i 's demographics or idiosyncratically.³⁹ Subsidies only affect utility through their effect on out-of-pocket payments.

Families choose the provider (or no provider) that maximizes their utility. The choice set family i can choose from depends on the market-year they live in, the age of their child, and income-based eligibility (in the case of Head Start). I define this choice set as J_{it} . The choice set includes private providers, public providers, and the outside option (or choosing to forgo formal care).⁴⁰ Market-years are discrete and non-overlapping.⁴¹

To simplify notation, we can divide utility into two components. First is the part of utility common to all families, δ_{jt} . For example, on average, families prefer higher-rated providers or providers with higher ξ_{jt} . Second is the part of utility specific to family i , μ_{ijt} . For example, eligible families may have a greater distaste for distance. Some families also need to travel longer distances to reach provider j or pay a smaller out-of-pocket payment because of a subsidy. If we assume ϵ_{ijt} is extreme value type one, the probability that family

³⁹During estimation, I will assume tastes vary by eligibility for subsidies and variation in taste for the inside option by the age of the child.

⁴⁰Utility for public providers is estimated the same as for private providers. I observe almost all characteristics for both public and private providers. The one exception are quality-ratings, which I do not observe for public 4-Ks. I treat public 4-Ks as if they have no quality-rating and include as a characteristic whether the provider is a public 4-K. Similarly, I only observe schedules for public 4-Ks, which I include as binary characteristics set to zero for private providers.

⁴¹In future work, the assumption of non-overlapping markets could be relaxed.

i chooses provider j is

$$Pr(i \text{ choose } j) = \frac{\exp(\delta_{jt} + \mu_{ijt}(sub_i, MaxSub_t))}{\sum_{j \in J_t} \exp(\delta_{jt} + \mu_{ijt}(sub_i, MaxSub_t))}.$$

Families are more likely to choose providers that are generally more desirable (e.g., have higher δ_{jt}) and where they prefer based on family-specific taste (e.g., have higher μ_{ijt}). Family i 's demand for provider j further depends on which other providers there are, as summarized in the denominator. If there are other, better, providers, family i is less likely to choose provider j .

Total enrollment at provider j relative to the overall size of the market is called market share (sh_{jt}). Enrollment is the sum of all families who actually choose j . While the econometrician does not directly observe μ_{ijt} , we can assume its distribution.

$$sh_{jt} = \int \frac{\exp(\delta_{jt} + \mu_{ijt}(sub_i, MaxSub_t))}{\sum_{j \in J_t} \exp((\delta_{jt} + \mu_{ijt}(sub_i, MaxSub_t)))} dF(\mu_{ijt})$$

Let's say we observe aggregate enrollment, but not individual family choices. We also do not observe where family i lives. However, we do observe the distribution of demographic types $\tau \in \mathcal{T}$ across census blocks in market t .⁴² Call the set of census blocks $\mathcal{L}_t$. I approximate the distance family i travels to provider j as the distance from the population-weighted centroid to the provider's address.

Say μ_{ijt} varies only by observable characteristics of provider j , family type τ (which encompasses subsidy status), idiosyncratic taste for observables, the census block the family lives at, and the maximum subsidy. Let w_{lt}^τ be the share of all families of type τ in the market t that live at block $l \in \mathcal{L}_t$. Let π_t^τ be the proportion of families in market t of type τ . We can then aggregate enrollment by summing over all types $\tau \in \mathcal{T}$ in each census block $l \in \mathcal{L}_t$ in the market, integrating out idiosyncratic taste for observables (e.g., random coefficients

⁴²I observe the distribution of families eligible or ineligible for subsidies across space. However, I do not observe where subsidy recipients live. I impute the location of subsidized families using data on where eligible families live across space and the number of subsidy recipients choosing a provider within market t .

on observables):

$$sh_{jt} = \sum_{\tau \in \mathcal{T}} \sum_{l \in \mathcal{L}(t)} w_{lt}^{\tau} \pi_t^{\tau} \int \frac{\exp(\delta_{jt} + \mu_{ijt,l}(\tau, MaxRate))}{\sum_{k \in \mathcal{J}(t)} \exp(\delta_{kt} + \mu_{ikt,l}(\tau, MaxRate))} dF(\mu_{ijt,l}(\tau)) \quad (5)$$

4.2 Supply

Family choices depend on the price, quality, and availability of providers. Families may prefer high-quality care, but high-quality care may not be available nearby. In this section, I discuss how providers adjust their prices or quality in response to changes in underlying demand. Observed quality ratings are updated annually in Wisconsin. Accordingly, I assume providers invest in quality, which increases the likelihood they earn a higher quality rating, but does not guarantee a particular rating. The provider only invests in quality if higher-quality providers can expect to earn a higher average profit in the next stage. After observing their quality rating, providers can set their prices to maximize profits.

First, providers observe their exogenous characteristics and the exogenous characteristics of other providers. Providers know the shape of demand, the maximum subsidy, and the proportion of families that will be subsidized in equilibrium, on average. With this knowledge, providers know what their expected profits are under each quality rating. Call this expected profit $\mathbb{E}\Pi(Qual_{jt} = q; X_{jt}^m, X_{jt})$, which depends on own characteristics X_{jt} and the characteristics of providers in the market X_{jt}^m . Nearby providers likely affect expected profits more than faraway providers, so X^m may differ by provider j in market-year jt , even within the same market. To fix ideas, say there are two quality ratings, “Lo” and “Hi.” I also normalize the marginal cost of effort to one. We can then say that the provider solves

$$\begin{aligned} \max_{e_{jt} \in \mathbb{R}^+} & \mathbb{E}\Pi(Qual_{jt} = Lo; X_{jt}^m, X_{jt}) * (1 - Pr(Qual_{jt} = Hi|e_{jt}, X_{jt})) \\ & + \mathbb{E}\Pi(Qual_{jt} = Hi; X_{jt}^m, X_{jt}) * Pr(Qual_{jt} = Hi|e_{jt}, X_{jt}) - e_{jt}, \end{aligned}$$

where $Pr(Qual_{jt} = q|e_{jt}, X_{jt})$ is the probability of earning a particular quality rating, given

effort expended and the characteristics of the firm. I assume throughout that the probability of earning a higher rating is increasing in effort.

The provider solves for optimal effort by solving first order conditions:

$$\left. \frac{\partial Pr(Qual_{jt} = Hi)}{\partial e_{jt}} \right|_{e^*} = \frac{1}{\mathbb{E}\Pi(Qual_{jt} = Hi|e^*) - \mathbb{E}\Pi(Qual_{jt} = Lo|e^*)},$$

where, for the sake of succinctness, I have dropped the notation for X_{jt} . Here, the provider considers both the marginal returns to effort on the likelihood they obtain a high-quality rating and the additional profit the provider can earn if they are rated high-quality. If we assume that marginal returns to effort on the likelihood of obtaining a higher rating are decreasing, the provider expends more effort on quality as the difference in expected profit from being high-quality relative to low-quality increases.⁴³ In Wisconsin, providers must be quality-rated to receive subsidies and all quality-rated providers must accept subsidies. The difference in expected profits between offering high- and low-quality then also encompasses differences in profits between accepting and rejecting subsidies.

After choosing the effort to expend, providers observe their own quality ratings, as well as the quality ratings of their competitors. Providers then set prices to maximize profits, given their knowledge of the shape of demand. Profit depends on enrollment sh_{jt} , the price the provider sets p_{jt} , and their marginal cost $mc_{jt}(Qual_{jt})$, which also depends on the provider's quality rating. We can say that profit represents the solution to the following maximization problem:

$$\Pi_{jt} = \max_{p_{jt} > 0} sh_{jt}(p_{jt} - mc_{jt}(Qual_{jt}))$$

⁴³Corner solutions ($e = 0$) are also possible. For example, if the provider actually expects *lower* profit from being high-quality, we should expect no effort. Similarly, if effort is unlikely to increase the probability of being high-quality, firms may expend no effort.

Solving for first order conditions, the provider sets price to marginal cost plus a mark-up:

$$\implies p_{jt} = mc_{jt}(Qual_{jt}) + \underbrace{\left| \frac{sh_{jt}}{\partial sh_{jt}} \right|}_{\text{mark-up}}.$$

Providers set higher prices if they have higher marginal costs, higher enrollment (market share), or if demand for provider j is more inelastic to provider j 's price.

The provider solves the first stage through backwards induction. The provider invests more in quality if the provider can charge a sufficiently higher mark-up compared to low-quality care.

4.3 Balanced Budget

The state must maintain a balanced budget: spending cannot exceed Y_t . Both the budget and the maximum subsidy are set exogenously. The state must then determine how many families to subsidize.

Some families are eligible for subsidies and others are not. Call the group of families eligible for subsidies $\mathcal{E}$. The state does not change eligibility criteria, so this group is fixed. Rather, the state subsidizes a certain proportion of families in $\mathcal{E}$, and places the remainder on a waitlist. In practice, the state solves for the equilibrium proportion of eligible families subsidized, $Pr(Sub)$, such that the budget balances. We can also write this problem as follows:

$$Y_t = Pr(Sub) \sum_{i \in \mathcal{I}(Elig)} \min\{MaxSub, p_{j(i,s=1),t}\}$$

Where the state pays the minimum of the maximum subsidy $MaxSub$ and the counterfactual price $p_{j(i,s=1),t}$ of the provider that family i would choose if they were subsidized.

Increasing the maximum subsidy does not *necessarily* reduce the number of families subsidized. Fixing family choices and the price and quality of care, increasing the maximum

subsidy reduces the number of families served if and only if there are some counterfactual prices $p_{j(i,s=1),t} > MaxSub$. These expensive providers would mechanically receive a larger subsidy if the state implemented a higher maximum subsidy.

4.4 Implications

Under what conditions can subsidy policy affect the quality of care families receive? First, I discuss how a family may be more likely to select a high-quality provider if they receive a subsidy. Second, I elaborate on the implications of subsidizing more families on the price and quality of care in the market. Third, I explain how, holding the proportion of families subsidized fixed, a higher maximum subsidy may affect a family's taste for quality and the resulting implications for the market equilibrium. Fourth, I consider the ambiguous effect of increasing the maximum subsidy under limited resources. Finally, I conclude by briefly discussing how public care may affect these results.

If a family is offered a subsidy,⁴⁴ their out-of-pocket payment falls by $\min\{p_{jt}, MaxSub\}$ for all formal providers accepting subsidies. The subsidy reduces the out-of-pocket payment more for providers with prices above the maximum subsidy. For all providers setting prices below the maximum subsidy, the family has an out-of-pocket payment of zero. Subsidized families then pay a smaller difference in out-of-pocket payment across providers. Providers are differentiated less by out-of-pocket payments and more by non-price characteristics, such as quality. If families are insensitive to the out-of-pocket payment, a subsidy will have a small effect on the family's choice. If families are sensitive to out-of-pocket payments, where families shift depends on their preferences for non-price characteristics and the available care nearby.

If more families are subsidized, the portion of demand coming from subsidized families, $w_{it}^{sub} \pi_t^{sub}$, increases and the portion of demand coming from eligible, but unsubsidized, families decreases. Prices increase for provider j in market-year t if families become less

⁴⁴Families are not obligated to accept a subsidy if they are offered one.

sensitive to the price provider j charges or enrollment sh_{jt} increases. Subsidized families are insensitive to marginal increases in the price below the maximum subsidy, since they pay nothing regardless. As more families are subsidized, providers can set prices closer to the maximum subsidy from below. Additionally, if more families are subsidized, demand for formal care increases, as does demand for providers with desirable, non-price characteristics increase. So mark-ups may increase more for providers with these desirable, non-price characteristics as more families are subsidized. If demand increases for high-quality care relative to demand for low-quality care, providers invest more in quality.

Let's now consider the case where the number of families subsidized is fixed, but the maximum subsidy increases from $MaxSub_1$ to $MaxSub_2$, where $MaxSub_1 < MaxSub_2$. The out-of-pocket payment to enroll at provider j falls by $\min\{\max\{0, p_{jt} - MaxSub_1\}, MaxSub_2 - MaxSub_1\}$, meaning that the out-of-pocket payment *only* falls for providers with $p_{jt} > MaxSub_1$ and falls the most for providers with $p_{jt} > MaxSub_2$. Increasing the maximum subsidy further limits the differences across providers, increasing demand for non-price characteristics.

If the maximum subsidy increases, only demand from subsidized families changes in response. If few families are subsidized, providers should not change their prices or quality in response. Subsidized families are indifferent to marginal price changes up to the new maximum subsidy, leading providers to increase prices from below the old maximum subsidy to below the new maximum subsidy. If a higher maximum subsidy also increases demand for non-price characteristics among subsidized families, prices may increase more for providers with desirable non-price characteristics. Providers may invest in quality if demand sufficiently shifts from low-quality to high-quality providers.

If resources are limited, increasing the maximum subsidy may reduce the number of families the state can serve. The extent to which the program contracts depends on three factors. First, payments to any provider with prices above the maximum subsidy are mechanically higher under the larger maximum subsidy. Second, subsidized families may

choose more expensive providers if they have a larger maximum subsidy, holding the prices of providers fixed. Third, providers may change their price in response to the higher maximum subsidy. Each of these factors stretch the resources of the program. But if the maximum subsidy reduces the number of families subsidized, the aggregate effect on demand is ambiguous. While the maximum subsidy may increase demand for non-price characteristics among subsidized families, there are now fewer subsidized families.

Public care may alter these dynamics by offering a free alternative care arrangement for families who would otherwise be subsidized. Additionally, public care may lower provider mark-ups by offering a competitive alternative. However, public care is not unlimited. In Wisconsin, the two most common public offerings are 4-K and Head Start. 4-K is only available to four-year olds and, in a handful of school districts, three-year olds. Head Start is only available to low-income children and may be located far away from some families. Subsidy policy changes may then have a larger effect where public care is scarce.

5 Estimation and Results

The extent to which offering a larger subsidy to more families compared to offering a smaller subsidy to fewer families leads to more eligible children enrolled in high-quality care depends on several parameters, including taste for price and quality, as well as how easy it is for a firm to obtain a higher quality rating. This section describes how I estimate the structural model and briefly discusses the estimates.

5.1 Demand

Demand estimation follows other work estimating spatial demand models of education markets (Bodere, 2023; Neilson, 2025), which extend the method from Berry et al. (1995).

Markets combine nearby school districts.⁴⁵ Since I do not observe family locations, I

⁴⁵Wisconsin’s geography lends itself to splitting the state into several regional markets. Besides Milwaukee and Madison, cities are small and relatively far apart from other cities. Here, markets center around the

divide each market into census blocks. Distance to a provider is taken as the distance from the centroid of the census block to the address of the provider.⁴⁶

The algorithm iteratively estimates demand in three steps. First, given a starting guess of μ_{ijt} , I estimate δ_{jt} by matching aggregate shares through a fixed-point problem. With δ_{jt} in hand, I can estimate ξ_{jt} as the portion of average demand unexplained by observables.

Second, I back out unobserved cost shifters based on implied mark-ups. Rearranging the price-setting equation:

$$mc_{jt} = p_{jt} + \frac{sh_{jt}}{\frac{\partial sh_{jt}}{\partial p_{jt}}}$$

I directly observe market share, sh_{jt} , and prices. $\frac{\partial sh_{jt}}{\partial p_{jt}}$ can be determined using estimated δ_{jt} , μ_{ijt} , and the marginal dis-utility of out-of-pocket payments β_i^{OOP} .⁴⁷ While I have now determined marginal costs, for counterfactuals we need to know the relationship between marginal costs and quality. Providers near capacity may also charge higher prices, biasing our estimates of marginal costs. With this in mind, I parameterize marginal cost as

$$mc_{jt} = \theta^X X_{jt}^S + \theta_t^Q Qual_{jt} + \theta^C \exp\left(\frac{sh_{jt}N_t}{Capacity_{jt}}\right) + \omega_{jt},$$

where ω_{jt} is the part of marginal cost unexplained by observables, $sh_{jt}N_t$ is total enrollment at provider j in market-year t , and $Capacity_{jt}$ is provider j 's capacity.

Although I now have an estimate of δ_{jt} and marginal cost, these estimates have relied on an initial guess of μ_{ijt} . In the final step, I estimate the i -specific parameters by Generalized Method of Moments. The GMM problem includes IV and micro-moments.

I instrument for the residual demand (ξ_{jt}) and supply (ω_{jt}) terms with exogenous own-

largest school districts and are contiguous. They are generally at least 40 miles wide. There are 37 markets in Wisconsin each year (2022-2025).

⁴⁶I use Haversine distance throughout.

⁴⁷Estimating price sensitivity requires an estimate of the marginal dis-utility of price rather than the marginal dis-utility of out-of-pocket payments. The marginal dis-utility of prices is the same as the dis-utility of out-of-pocket payments β_i^{OOP} for unsubsidized families. For subsidized families, the marginal dis-utility of price is zero below the maximum subsidy and equals β_i^{OOP} above the maximum subsidy.

characteristics, spillovers from a child care worker wage subsidy program⁴⁸, distance to income verification offices, and BLP-style instruments that measure the availability of different kinds of providers in the market. Let Z^D be the set of demand-side instruments and Z^S the set of supply-side instruments. These IV moments, where θ is the full set of parameters, are:

$$g^{IV}(\theta) = \frac{1}{N} \begin{bmatrix} Z^D & 0 \\ 0 & Z^S \end{bmatrix} \begin{bmatrix} \xi(\theta) \\ \omega(\theta) \end{bmatrix}$$

I stack micro-moments to the GMM problem. First, I match average quality, conditional on enrollment, by type τ ⁴⁹ in market t . From the data, let N_t^{priv} be the number of families choosing a private provider in the data and $q_{j(i)}$ the observed quality rating of the provider that family i chooses. Let $\mathcal{I}(\tau, t)$ be the observed families of type τ in market t . The estimates of δ_j, μ_{ijt} then imply a share of families of type τ who choose a provider of quality rating q , conditional on the family choosing a private provider. For the GMM problem, we can write the matched problem as follows. Let $s_{ljt, \tau}^{priv}$ be the share of families of type τ living at census block l enrolled at provider j relative to all *private* enrollment – excluding enrollment at public providers or the outside option, which by construction have no quality rating. The implied share depends on our current estimates of the parameters, which collectively I refer to here as θ . Let w_l^τ be the proportion of families of type τ that live at census block l . We can then add to the GMM problem:

$$g_{\tau, t}^{MM, q} = \frac{1}{N_t^{priv}} \sum_{i \in \mathcal{I}(\tau, t)} q_{j(i)} - \sum_{l \in \mathcal{L}_\square} w_l^\tau \sum_{j \in \mathcal{J}_\square} s_{ljt, \tau}^{priv}(\theta) q_{jt}$$

I match similar moments for average out-of-pocket payment paid by type.

Additionally, I use micro-data on public 4-K enrollment to match the fraction of all public 4-K enrollment in a market (N_t^{pub}) from children eligible for Free or Reduced Price

⁴⁸This program operated under Child Care Counts. Providers with higher quality ratings could earn a higher per-worker wage subsidy. Providers in markets with more high-quality providers or in years with higher wage subsidies are subject to higher average wage subsidies, conditional on their own quality.

⁴⁹I observe enrollment by subsidy status, so τ here is either subsidized or unsubsidized

Lunch. Similarly to the private moments, s_{jt}^{pub} is the model-implied share of families enrolled in public 4-K:

$$g_{\tau,t}^{MM,Elig} = \frac{1}{N_t^{pub}} \sum_{i \in \mathcal{I}(\tau,t)} Elig_i - \sum_{l \in \mathcal{L}_{\sqcup}} w_l^{\tau} \sum_{j \in \mathcal{J}_{\sqcup}} s_{jt}^{pub}(\boldsymbol{\theta}) Elig_i$$

Lastly, I match moments from national survey data on distance traveled⁵⁰ and the share of private enrollment coming from children of different ages.

I then choose β_i to maximize the GMM objective function. With these updated parameters in hand, I repeat the algorithm until convergence.

5.2 Quality Choice

After I estimate demand, I estimate the parameters in the quality-choice stage. Say there are $\mathcal{Q} = \{1, 2, 3, 4, 5\}$ possible discrete quality levels. I parameterize the probability of becoming quality $q \in \mathcal{Q}$ as:

$$\begin{aligned} 1\{Q > q | Q > q - 1\} &\sim \text{Logit}(\beta_1^q X_j \ln(e) + \beta_2^q X_j, 1) & (\forall q \in \{2, 3, 4\}) \\ 1\{Q > 1\} &\sim \text{Logit}(\beta_1^1 X_j \ln(e) + \beta_2^1 X_j, 1) \end{aligned}$$

In my estimation, only whether the private provider is a center or a family provider appears in X_j . Because markets can be large, I follow the dynamic programming literature and approximate expected profits.⁵¹ I simulate profits across many quality choice mixes, then predict average profits given own quality choice based on observable features.⁵²

Based on a starting β^0 and $\mathbb{E}\pi$, I determine the optimal e using grid-search. After determining e , I can estimate the distribution of a summary measure of the actions taken by

⁵⁰Specifically, the share of families, by urbanicity and subsidy eligibility, who travel zero to one mile, one to three miles, or three to eight miles to reach their primary care provider.

⁵¹As in Bodere; Sweeting.

⁵²Examples include measures of local demographics and local competitor characteristics.

nearby providers (closest five or within five miles), a_N and a summary measure of the actions taken by all providers in the market a_M .⁵³ With that distribution in hand, I estimate:

$$\mathbb{E}\pi_k = \int_{a_M, a_N} \pi_k(a_M, a_N) * Pr(a_M, a_N)$$

Using the updated expected profits, I re-estimate e . I continue until $|e^n - e^{n+1}|$ is less than a given tolerance. Once I have e in hand, I choose β to best match predicted and actual quality in the sample using maximum likelihood estimation. I repeat the entire iterative process until β also converges.

5.3 Estimates⁵⁴

5.3.1 Demand and Marginal Costs

Table 11 reports the demand estimates. I allow for provider characteristics such as its type (e.g., center, family, public fee-based⁵⁵, public 4-K, or Head Start), the age of children served, how many years the provider has operated for, and the quality rating. For public 4-Ks, I allow for demand for the public 4-K to vary by the operating schedule of the 4-K provider.

I also allow for demand to vary across families in five ways. First, I allow demand for a given provider to depend on the distance the family is from the provider. Because I do not observe family addresses but do know the distribution of families across space by income, I take distance traveled to be the Haversine distance from the centroid of the family’s census block to the provider. I allow taste for distance to also vary by whether the family lives in Milwaukee, Wisconsin’s densest city. Second, I allow the out-of-pocket payment to depend on the subsidy status of the family.⁵⁶ Receiving a subsidy affects demand only through its

⁵³The summary measure I use here is the average quality rating. The distribution is either measured exactly in the case of small markets, via Gauss-Hermite quadrature in the case of markets with at least thirty providers, and via Monte Carlo simulation for “medium”-sized markets.

⁵⁴These results are preliminary. Please refer to the most up-to-date version of this paper.

⁵⁵These providers operate in public schools and require families to pay a fee. I treat these providers throughout as if they are not price-setters.

⁵⁶I do not observe the exact address of subsidized families. For sake of estimating the model, I assume

effect on the out-of-pocket payment. Third, families eligible for subsidies have different tastes for all provider characteristics than ineligible families, capturing variation in preferences due to income. Fourth, I allow individual variation in taste for the highest-quality rating. Fifth, I allow the age of the child to affect taste for formal care and the options available.⁵⁷

I find that demand for formal care is highest for families with three-year-olds and for ineligible families. While all families have a significant distaste for the out-of-pocket payment they pay for care, I find that ineligible families are more sensitive to their out-of-pocket payment. Eligible families and families living in Milwaukee are also significantly more sensitive to the distance they need to travel to their provider.

Conditional on a provider being rated, families generally prefer higher-rated providers. Eligible families prefer middling-quality providers, all else equal. Families prefer unrated providers to low-rated providers, perhaps reflecting beliefs that unrated providers may be higher quality, but have strategically chosen not to be rated. I also find significant variation in taste for the highest-quality rating, implying that quality is more important for some families regardless of demographics.

5.3.2 Marginal Costs

Table 12 describes the marginal cost estimates. Marginal cost varies with the provider's quality rating, ages served, years in operation, facility type, and average county wages. Providers may set higher prices as they approach capacity, biasing estimation of marginal costs, which are based on estimated mark-ups. To address this bias, I allow for marginal costs to increase as the provider approaches capacity.

Marginal costs are generally increasing in quality. In particular, the highest-quality providers have significantly higher marginal costs than other quality ratings. Marginal costs are falling in years in operation. Centers pay lower marginal costs than family providers.

eligible families have an equal chance of being subsidized across a given market.

⁵⁷If a provider does not accept an age group, I assume the provider gives families with children of that age no utility, regardless of the provider's other characteristics.

Provider costs are decreasing with the youngest age served after about 19 months and insignificantly decreasing with the oldest age served. I find evidence that providers set higher prices as they approach capacity.

5.3.3 Quality Stage

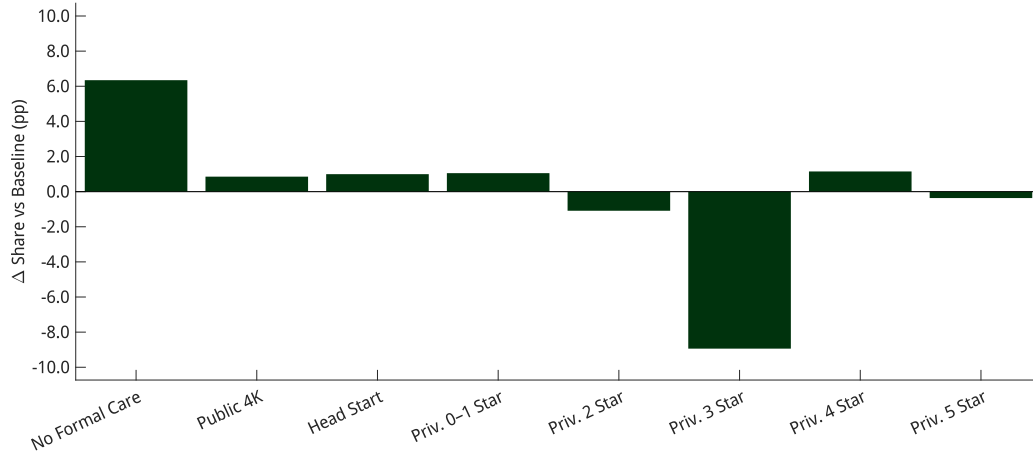
Family providers and centers are rated under similar, but slightly different methodologies, so I estimate the quality-transition stage separately for family providers and centers. I find that centers can more easily achieve a rating of two stars compared to family providers. However, family providers must expend less effort to reach higher ratings compared to centers.

6 Counterfactuals

I evaluate the 2022 Wisconsin reform, which raised the maximum subsidy from the 35th percentile of local prices to the 80th percentile. I compare the effect of the reform relative to using the same total budget to subsidize more families under the previously smaller maximum subsidy. I decompose the difference between the two policies into three channels: sorting (holding prices and quality fixed), endogenous prices, and endogenous quality. The results below focus on the effects on children eligible for subsidies under age five. [Figure 5](#) shows the overall effect of the higher maximum subsidy on enrollment, relative to the smaller subsidy offered to more families. I find that the largest effect on enrollment is to push eligible children out of formal care. 6.3pp fewer eligible children enroll in formal care, an 18% decline. Most of these children come from 3-star providers.

The decline in enrollment in formal care comes predominantly from sorting and price increases. The right-most column in [Figure 6](#) shows the effect of the higher maximum subsidy, relative to the smaller maximum subsidy offered to more families, on choosing to not enroll in formal care. The first column shows the relative effect from changes in family choices alone. Subsidizing fewer families explains 38% of the decline in formal care

Figure 5: Enrollment in Formal Care Falls Under the Larger Subsidy



Shows changes in enrollment among eligible children under age five. Changes compare enrollment under the maximum-subsidy reform relative to offering a smaller subsidy to more families. Columns sum to zero.

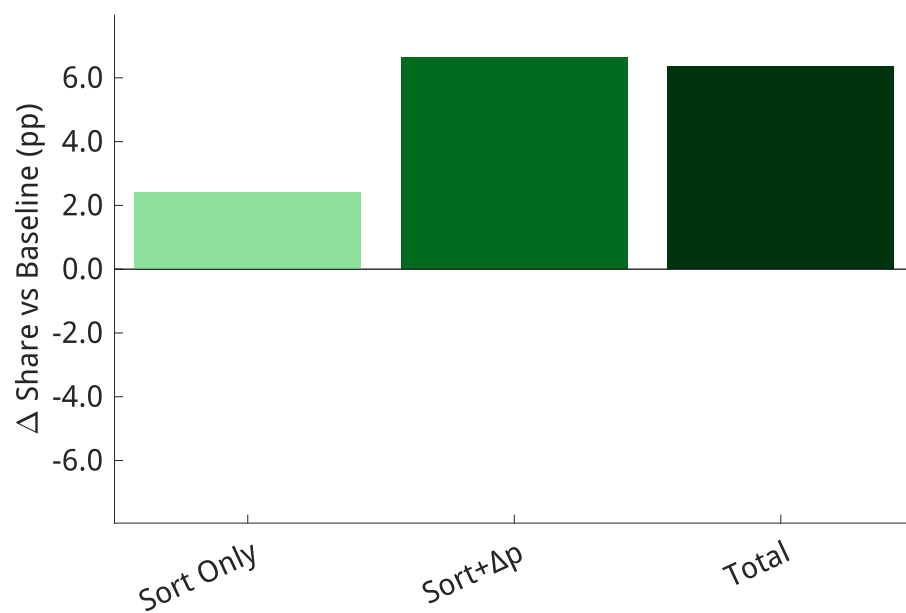
enrollment. The remainder is explained by providers increasing prices, as evidenced by the second column. Indeed, as shown in [Figure 7](#), providers raise sticker prices by 9% under the higher maximum subsidy.

Public alternatives absorb some children who would otherwise have been subsidized. Public 4-K and Head Start enrollment increases by 18% under the reform. [Figure 8](#) shows how this effect can be decomposed into the effects of a higher maximum subsidy and fewer families subsidized. The effect is predominantly from families losing the subsidy rather than the maximum subsidy itself, suggesting that the subsidy crowds out public enrollment in part.

Enrollment in high-quality⁵⁸ private care increases. Under the higher maximum subsidy, 13% more eligible children enroll at high-quality private providers. [Figure 9](#) decomposes the change into three channels. Sorting explains about half of the increase. With the larger maximum subsidy, subsidized families choose higher-quality providers. The remainder is

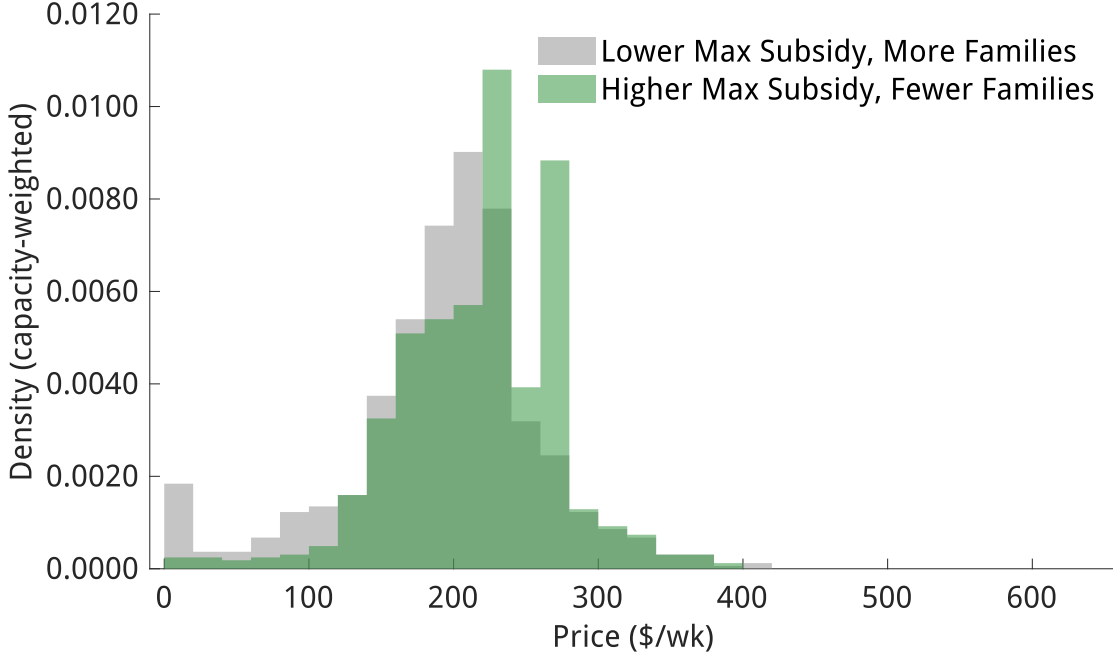
⁵⁸Defined here as 4+ stars

Figure 6: Prices Explain Most of Increase in Families Opting Out of Formal Care



Change in formal-care enrollment by eligible families, decomposed into family choices, price changes, and quality changes.

Figure 7: Sticker Prices Increase Under Larger Subsidy



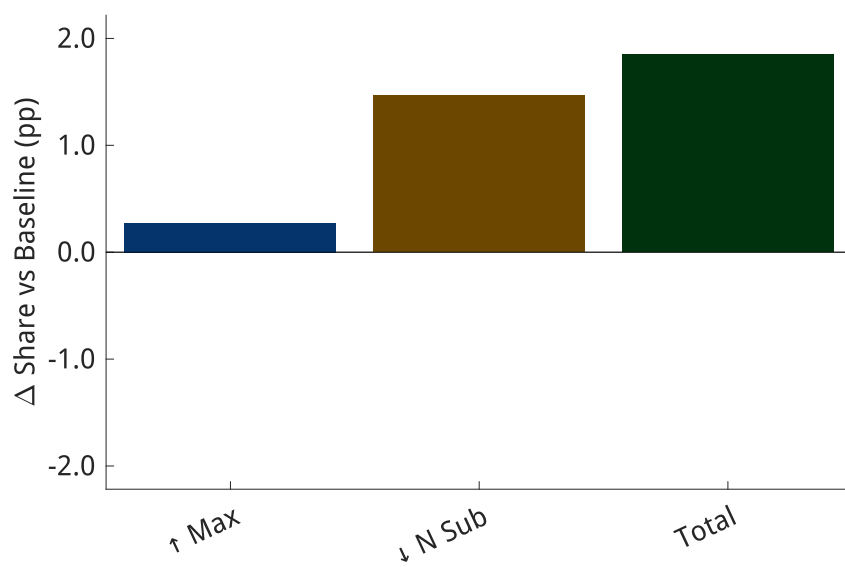
Distribution of provider sticker prices under the maximum-subsidy reform relative to subsidizing more families.

explained by providers investing in quality. I find that, in particular, the larger subsidy induces unrated providers to increase their ratings, as shown in [Figure 10](#).

7 Conclusion

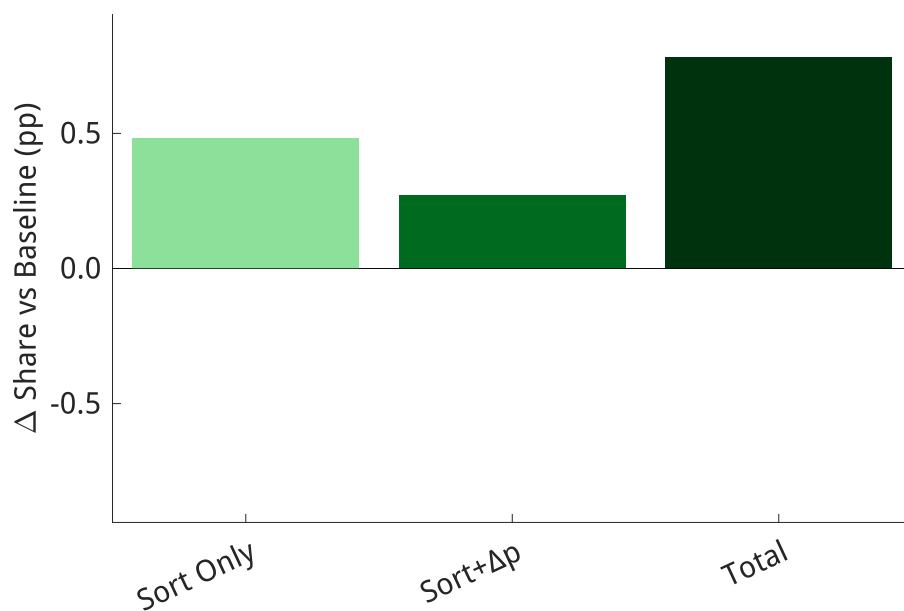
Under a fixed budget, state child care subsidy programs must ration. If states increase the maximum subsidy, they must ration further. Using quasi-experimental variation, I demonstrated that providers change their price and quality in response to changes in how many families receive subsidies and by how much. To determine the budget-neutral counterfactual and disentangle changes in family choices from equilibrium price and quality changes, I estimated a model of the child care market. I found that the maximum subsidy increased subsidized families' demand for high-quality care, leading to both higher prices and quality. Families who are eligible, but are unable to access the subsidy, are priced out of formal care.

Figure 8: Public Enrollment Increases Under Maximum Subsidy Because of Reduced Crowd-Out



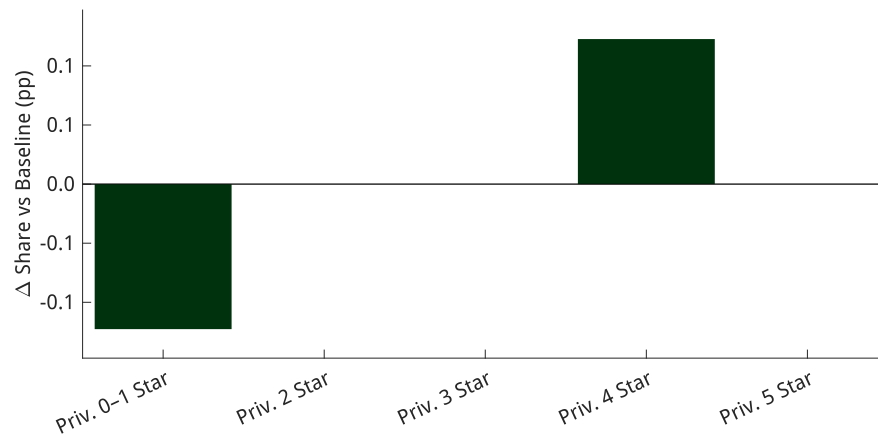
The effects of (1) offering a higher maximum subsidy offered to more families, (2) offering the smaller subsidy to fewer families, and (3) offering the larger subsidy to fewer families, relative to offering a smaller subsidy to more families

Figure 9: High-Quality Private Enrollment Increases Because of Sorting, Higher Private Quality



Change in high-quality private-care enrollment from moving from a smaller maximum subsidy to a larger maximum subsidy if (1) provider prices and quality are held fixed, (2) provider quality is held fixed, and (3) prices and quality can both change.

Figure 10: Quality is Higher Under Larger Maximum Subsidy



Overall change in the distribution of provider quality from the smaller maximum subsidy to the larger maximum subsidy.

While some unsubsidized families move to public care, more families drop out of formal care.

To place these results in context, Head Start has the largest effect on the development of children who would otherwise not have attended any formal care ([Feller, Grindal, Miratrix and Page, 2016](#); [Kline and Walters, 2016](#)). The large reduction in formal care use may then be especially harmful for children's long-run outcomes. Additionally, while public care enrollment is preferred to no formal care enrollment, public care is not free for the state to provide. Incorporating these costs would further reduce the number of families the state could serve. Access to affordable formal care may also allow low-income parents to work and attend school. Increasing family income may reduce fiscal costs through reduced use of other benefits. Alternatively, if the federal government wishes to increase access to high-quality care through higher maximum subsidies, it should offer the states enough funding to achieve this goal.

This paper has limitations. I do not consider the endogeneity of costs of providing care. As providers improve their quality, demand for child care workers may increase, increasing wages. Subsidies would then have a smaller effect on the quality offered by providers.

Additionally, I do not explicitly model entry, exit, or capacity constraints, instead treating the lack of capacity constraints as a proxy for free entry. Entry may be costly and I do not model entry into new locations. Capacity constraints could push providers to charge higher prices and limit the effectiveness of expanding access to subsidies in the absence of entry. Both the role of an endogenous child care labor market and capacity constraints are the subject of ongoing work.

In this paper, I demonstrate that density of program participation matters when determining the price and quality of private services. Other programs, like housing or health care, may be similarly affected. I also show that cost-effectiveness requires evaluating the opportunity cost of funds. Many state and local programs rely on federal grants that are earmarked for certain purposes. Understanding how these programs can work better requires evaluating alternative program designs within a given budget.

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A Sample Descriptives

Table 9: Model-Estimation Sample vs. All Licensed Providers

	(1) WI Model (2022-2025)	(2) WI Lic. + Publics (2022-2025)	(3) Inc. Missing Data (2022-2025)
<i>Panel A: Full Sample (Enrollment-Weighted)</i>			
Family provider	0.056	0.056	0.060
Public: Fee-Based	0.030	0.029	0.029
Public 4K	0.306	0.271	0.304
Head Start	0.061	0.055	0.060
<i>Panel B: Private Sample (Enrollment-Weighted)</i>			
Quality rating	3.10	2.89	3.03
Sticker price, mo. (2017)	849.40	815.90	849.40
OOP, subsidized (mo., 2017)	70.83	70.35	71.42
Cap.-constrained (lic. only)	0.348	0.383	0.352
Min. age accepted (months)	5.5	11.5	5.5
Max. age accepted (months)	144.1	143.6	143.8
Serves school-age only (5+)	0.000	0.068	0.000
Years of operation	15.2	15.2	15.0
Total est. subsidized enrollment	98,686	120,780	107,420
Total est. enrollment	346,874	423,500	355,737
<i>Panel C: Public 4K Only (Raw Means)</i>			
Full day, full week	0.324	0.324	0.324
Half day, full week	0.287	0.287	0.287
Full day, half week	0.100	0.100	0.100
Half day, 4 days/week	0.289	0.289	0.289
Charter school	0.052	0.066	0.066
Offers K3	0.068	0.067	0.067
Share FRPL-eligible	0.376	0.377	0.377

This table compares Wisconsin child care providers across three overlapping samples for 2022-2025. Column (1), the WI Model sample, includes Wisconsin licensed child care providers and Public 4Ks that are year-round and full-time; not a licensed camp or certified family provider; accept children under age five; report enrollment; linked to an address; and report prices. Column (2), WI Lic. + Pub., includes all Wisconsin licensed providers plus all Public 4K schools, without any additional filters. Column (3) includes providers that are otherwise in-sample but are dropped for missing enrollment, address, or price. Panels A and B weight by each provider's reported enrollment. Providers with missing enrollment do not contribute to those panels. Panel B is restricted to the private sample, which excludes Public 4Ks, Head Start, and public fee-based providers. Panel C is restricted to Public 4Ks and reports unweighted means across those schools. "Family providers" offer child care from a residential facility. "Public: Fee-Based" is a licensed provider operating from a public school. These providers charge fees, unlike public 4Ks. "Quality Rating" is the provider's YoungStar quality rating on a 1-5 scale (0 = not rated). "Sticker Price" is the monthly price an unsubsidized family would pay to attend. "OOP, subsidized" is the estimated monthly out-of-pocket payment for subsidized children, weighted by subsidized enrollment. Monthly prices and OOPs are for three-year-olds, expressed in 2017 dollars. "Cap.-constrained" is the share of licensed providers whose estimated enrollment meets their licensed capacity. Panel C describes Public 4K program schedules. Full-day is generally a full school-day, rather than a full work-day. Half-week programs generally operate two to three days a week. "Share FRPL-eligible" is the share of Public 4K children eligible for federal free or reduced-price lunch.

Table 10: Model-Estimation Sample vs. Stylized Facts Sample: Private Providers

	(1) WI Model Sample (2022-2025)	(2) WI Licensed (2022-2025)	(3) WI Sample (2017-2020)	(4) PA Sample (2017-2020)
Family provider	0.064	0.069	0.102	0.050
Public: Fee-Based	0.082	0.105	0.000	0.000
Head Start	0.112	0.109	0.000	0.000
High-quality	0.315	0.259	0.280	0.335
Sticker price, mo. (\$2017)	760.15	718.00	909.37	823.55
Avg sticker price if > 0	856.45	811.92	909.37	823.55
No subsidized enrollees	0.135	0.225	0.180	0.023
Serves school-age only (5+)	0.000	0.068	0.000	0.000
Total licensed seats	524,150	715,882	494,491	1,272,832
Number of providers	10,672	17,874	13,025	22,077

This table compares private child care providers across four sub-samples: the Wisconsin sample used in the structural model; all licensed WI private providers from 2022-2025; and the Wisconsin and Pennsylvania sub-samples used for the stylized facts. All four columns exclude Public 4Ks. All main-body rows report capacity-weighted means “Total seats” is the unweighted sum of licensed capacity across all providers in the column, and “Total providers (N)” is the unweighted provider count. “Family providers” offer child care services from a residential facility. “Public: Fee-Based” is a licensed provider operating from a public school. These providers charge fees for participation, unlike public 4Ks. “High quality” uses state-specific thresholds on the quality rating that require scoring an average of four on the Environment Rating Scale. “Sticker price, mo.” is the monthly price for a three-year-old in 2017 dollars; “Avg sticker price if > 0” restricts the mean to providers charging a positive price. “No subsidized enrollees” is the share of providers with zero subsidized enrollees, where such data was available.

B Model Estimates

Table 11: Demand-Side Parameter Estimates

	Est.		Est.
2 Stars	−6.2114 (2.5969)	4 Stars × Elig	−0.5501 (0.1874)
3 Stars	−10.9830 (2.5846)	5 Stars × Elig	−7.6373 (0.2770)
4 Stars	−2.4627 (3.0677)	Out-of-Pocket × Elig	−4.5385 (0.0974)
5 Stars	3.4796 (2.8199)	Out-of-Pocket × Inelig	−2.2864 (0.0513)
Center-Based	−0.5531 (1.1664)	Center × Elig	14.9996 (0.3372)
Capacity	0.1022 (0.0118)	Elig × Public 4K	−10.4724 (0.4129)
Youngest Age	0.2263 (0.0412)	Elig × Pub 4K, FDHW	−2.7503 (2.0321)
Oldest Age	−0.1254 (0.0191)	Elig × Pub 4K, HDFW	−6.3655 (0.5355)
Years in Op.	−0.7484 (0.0481)	Elig × Pub 4K, HD4D	−6.5367 (0.4403)
Inside Option	125.8478 (4.7205)	Elig × Pub 4K Fee	6.6290 (0.2354)
Head Start	−68.4632 (3.2737)	Distance × Elig	−14.9992 (0.3695)
Public 4K	−35.6361 (4.3703)	Distance × Inelig	−0.2213 (0.2091)
Pub 4K, FDHW	77.5105 (6.3138)	Distance Squared	−10.1714 (0.2370)
Pub 4K, HDFW	5.0965 (2.9320)	Dist × Milw × Elig	−0.0648 (1.1175)
Pub 4K, HD4D	8.7906 (3.2249)	Dist × Milw × Inelig	−2.4916 (0.6755)
Public Fee-Based	−26.0161 (2.6787)	Age 0–2 × Inside	8.3895 (0.2930)
Eligible	−3.4020 (0.2327)	Age 3 × Inside	18.2254 (0.5050)
2 Stars × Elig	2.1306 (0.1822)	Age 4 × Inside	0.7797 (0.1912)
3 Stars × Elig	10.9624 (0.2687)	σ (High-Quality)	9.3260 (0.2312)

Star coefficients are relative to a 1-star baseline. Inside Option applies to any formal provider. Public 4K variants FDHW, HDFW, and HD4D denote the full-day/half-week, half-day/full-week, and half-day/four-day schedules. Public Fee-Based refer to public fee-based providers, not limited to 4-K. Out-of-Pocket is the family’s out-of-pocket price. Distance is Haversine distance from the centroid of the family’s tract to the provider. Milw denotes Milwaukee County. Age 0–2, 3, and 4 reflect differences in taste for the inside good for difference age groups. σ is the standard deviation of the random coefficient on high-quality (5 star) providers. Standard errors in parentheses.

Table 12: Marginal Cost Parameter Estimates

	Est.		Est.
2 Stars	0.3910 (0.1754)	Oldest Age Served	−0.0022 (0.0113)
3 Stars	0.2588 (0.1840)	Oldest Age Squared	0.0000 (0.0000)
4 Stars	0.4551 (0.2261)	Years in Op.	−0.0214 (0.0114)
5 Stars	1.9998 (0.1741)	Years in Op. Squared	0.0011 (0.0003)
Local Wages	−0.0093 (0.0026)	Center-Based	−0.7606 (0.1038)
Youngest Age	0.0347 (0.0152)	exp(Utilization)	0.1880 (0.0643)
Youngest Age Sq.	−0.0009 (0.0004)		

Star coefficients are relative to a 1-star baseline. Local Wages is the county-level average child-care worker wage. Years in Op. is the number of years the provider has had their license. Utilization is the effect of operating near capacity on marginal costs. Standard errors in parentheses.

Table 13: Quality-Transition Estimates

Quality Transition	Slope on ln(effort)		Intercept	
	Family	Center	Family	Center
Base → 2 Stars	0.149	0.487	4.194	10.000
2 Stars → 3 Stars	0.020	0.001	—	—
3 Stars → 4 Stars	0.070	0.038	—	—
4 Stars → 5 Stars	0.023	0.001	—	—

Family providers operate from residential facilities while centers operate from commercial facilities. Each are assessed under similar, but different, standards. Intercepts are estimated only for the base → 2-Star transition. Providers must expend effort to reach a higher rating. Investing in quality is assumed to always have a positive effect on reaching higher quality-ratings.

C Optimal Effort Under Multiple Discrete Quality Types

Define

$$\begin{aligned}\frac{dP^i}{de} &\equiv \Pr(Q > i | Q > i - 1) \\ \pi_i &\equiv \mathbb{E}[\pi | Q = i]\end{aligned}$$

Let's say there are five quality ratings. The provider solves the following effort problem:

$$\begin{aligned}\max_{e_j} & \pi_1 \Pr(Q = 1) + \pi_2 \Pr(Q = 2) + \pi_3 \Pr(Q = 3) + \pi_4 \Pr(Q = 4) + \pi_5 \Pr(Q = 5) - e \\ \max_{e_j} & \pi_1(1 - \Pr(Q > 1)) + \pi_2(\Pr(Q > 1) - \Pr(Q > 2)) + \pi_3(\Pr(Q > 2) - \Pr(Q > 3)) \\ & + \pi_4(\Pr(Q > 3) - \Pr(Q > 4)) + \pi_5 \Pr(Q > 4 | Q > 3) \Pr(Q > 3) - e \\ \max_{e_j} & \pi_1 + \Pr(Q > 1)(\pi_2 - \pi_1) + \Pr(Q > 2 | Q > 1) \Pr(Q > 1)(\pi_3 - \pi_2) \\ & + \Pr(Q > 3 | Q > 2) \Pr(Q > 2 | Q > 1) \Pr(Q > 1)(\pi_4 - \pi_3) \\ & + \Pr(Q > 4 | Q > 3) \Pr(Q > 3 | Q > 2) \Pr(Q > 2 | Q > 1) \Pr(Q > 1)(\pi_5 - \pi_4) - e\end{aligned}$$

The first order condition is solved by finding the e such that the following holds:

$$\begin{aligned}1 = & \frac{dP^1}{de}(\pi_2 - \pi_1) + (P^2 \frac{dP^1}{de} + \frac{dP^2}{de} P^1)(\pi_3 - \pi_2) \\ & + (\frac{dP^3}{de} P^2 P^1 + P^3 \frac{dP^2}{de} P^1 + P^3 P^2 \frac{dP^1}{de})(\pi_4 - \pi_3) \\ & + (\frac{dP^4}{de} P^3 P^2 P^1 + P^4 \frac{dP^3}{de} P^2 P^1 + P^4 P^3 \frac{dP^2}{de} P^1 + P^4 P^3 P^2 \frac{dP^1}{de})(\pi_5 - \pi_4)\end{aligned}$$

Say that the returns to effort on achieving a higher rating are decreasing. Qualitatively, firms invest more effort as the marginal returns of achieving a higher rating increase. Differently from the two-level case, providers expend less effort if $\pi_5 > \pi_4$ and $\pi_{i < 5} = \pi_{i-1}$ than if $\pi_5 > \pi_4 > \pi_3 > \pi_2 > \pi_1$.